

From Source to Execution: Design and Implementation of a Statically-Typed Programming Language with Type Inference

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ABSTRACT

This report presents the design and implementation of a small, statically-typed interpreted programming language realised through a four-stage compiler pipeline. The language provides four primitive types — Integer, Float, Boolean, and String — and supports arithmetic and equality expressions, variable assignment with type inference from first use, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` function. Types are inferred from program context rather than declared explicitly, combining static type safety with a declaration-free syntax. The pipeline consists of a hand-written lexer, a recursive-descent parser that constructs an abstract syntax tree, a static type checker that performs inference and rejects ill-typed programs before execution, and a tree-walking interpreter. Both a command-line runner and an interactive desktop interface expose each pipeline stage for inspection. The report documents the formal grammar, the type inference algorithm, the system architecture, and the automated test suite of 54 tests that validates the complete implementation.

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CHAPTER 1

INTRODUCTION

Programming languages define the formal contract between a programmer’s intent and machine execution. Building a language processor from source text through to evaluated output requires a coherent pipeline of components, each of which must correctly transform its input before passing it downstream. Lexical analysis identifies atomic tokens; parsing imposes hierarchical structure; static type checking validates semantic correctness before execution; and interpretation evaluates the validated program. When these stages are connected cleanly, errors are caught at the earliest possible point, and the overall system remains tractable to reason about, test, and extend.

This report presents the design and implementation of a small statically-typed interpreted language that realises this pipeline in full. The language supports four primitive types — Integer, Float, Boolean, and String — and provides arithmetic and equality expressions, assignment, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` statement. Types are inferred from program context rather than declared explicitly, combining the safety of static typing with the conciseness of a declaration-free syntax. The implementation spans three principal components developed in pipeline order: the lexer and symbol table, the recursive-descent parser and abstract syntax tree, and the type checker with interpreter. Each component was completed before the next was begun, since each stage consumes the structured output of its predecessor.

The implemented system can be invoked from the command line or through an interactive desktop interface, both of which surface the four pipeline stages — tokens, AST, inferred type table, and execution output — for inspection. An automated regression suite of 54 tests verifies correctness across all stages and language features.

The remainder of this report is organised as follows. Chapter 2 presents the language design, token specification, grammar, and operator precedence rules. Chapters 3 and 4 cover the front-end implementation in pipeline order: lexical analysis and the symbol table, then recursive-descent parsing and the AST. Chapter 5 explains the type inference algorithm and the semantic rules used by the checker. Chapter 6 presents the system architecture and implementation,

including the end-to-end pipeline, project layout, and technology stack. Chapter 7 discusses testing and verification, and Chapter 8 concludes the report.

CHAPTER 2

LANGUAGE DESIGN AND GRAMMAR

2.1 Types

The language provides four primitive types, chosen to cover the core domains of numeric computation, logical control flow, and text manipulation while keeping the type system tractable and the semantics well-defined.

2.1.1 *Integer*

The Integer type represents whole numbers such as 0, 7, and 42. Integer literals are written as one or more decimal digits without quotation marks or a decimal point.

2.1.2 *Float*

The Float type represents real-valued numbers written with a decimal point, such as 3.14 or 20.5. Float values are used in arithmetic expressions and also arise as the result of any division. Division always produces a Float even when both operands are Integer: for example, `10 / 2` evaluates to `5.0 : Float`. This is an explicit design rule enforced in the type checker.

2.1.3 *Boolean*

The Boolean type represents logical truth values. The two Boolean literals are `true` and `false`. Boolean values are used in conditions and as results of comparison expressions.

2.1.4 *String*

The String type represents sequences of characters enclosed in double quotes, for example `"hello"`. Strings can be assigned to variables and printed through the built-in `print()` function. The surrounding quotes are stripped by the parser when the `Literal` node is created, so the value stored internally is the bare character sequence without surrounding quotes. At runtime, the built-in `print()` function re-adds double quotes around string values, so `print("plc")` outputs `"plc" : String`.

2.1.5 Void

Although the language exposes exactly four programmer-visible types, the implementation introduces a fifth value, `Void`, within the `LanguageType` enumeration. Its sole purpose is to provide a well-defined return type for functions that contain no `return` statement, allowing the symbol table to record a consistent entry for every function regardless of whether it produces a value. `Void` cannot appear in source code and is never visible to the programmer; it is a bookkeeping device used internally by the type checker.

2.2 Static Typing

The language employs static typing, meaning that type errors are detected at type-checking time rather than at runtime. No explicit variable type declarations are required. Instead, the type checker infers types from literals, assignments, operations, function calls, and return statements. Once a variable type is established, later assignments must remain consistent with that type. For example, the following program is rejected at the type-checking stage before any execution occurs:

```
1 x = 5;
2 x = "hello";    (* TypeError: Cannot assign String to variable 'x' of
   type Integer *)
```

This contrasts with dynamically-typed languages, where the same program would run without error until (or unless) a type-dependent operation was attempted.

2.3 Token Specification

Table 2.1 summarises the major token classes used by the language.

Table 2.1. Token Specification

Token Category	Example Lexemes
INT_LITERAL	0, 10, 42
FLOAT_LITERAL	3.14, 20.5
BOOL_LITERAL	true, false
STRING_LITERAL	"hello"
IDENTIFIER	variable and function names such as <code>x</code> , <code>add</code>
Arithmetic operators	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>
Comparison operators	<code>==</code> , <code>!=</code>
Assignment operator	<code>=</code>
Keywords	<code>if</code> , <code>else</code> , <code>while</code> , <code>def</code> , <code>return</code> , <code>print</code>
Delimiters	<code>(</code> <code>)</code> <code>{</code> <code>}</code> <code>,</code> <code>;</code>

2.4 Context-Free Grammar

The language grammar can be described by the context-free grammar $G = (V_N, V_T, P, S)$, where V_N is the set of non-terminals, V_T is the set of terminals, P is the set of production rules, and S is the start symbol program. The grammar is implemented through recursive-descent parsing functions.

2.4.1 Program Structure

```
1 program    -> statement* EOF
2 statement  -> assignment
3           | if_stmt
4           | while_stmt
5           | func_def
6           | print_stmt
7           | return_stmt
8           | expr ';' ;
```

2.4.2 Arithmetic Expressions

```
1 expr       -> term (( '+' | '-' ) term)*
2 term       -> factor (( '*' | '/' ) factor)*
3 factor     -> '-' factor
4           | INT_LITERAL
5           | FLOAT_LITERAL
6           | STRING_LITERAL
7           | BOOL_LITERAL
8           | IDENTIFIER
9           | IDENTIFIER '(' args? ')',
10          | '(' expr ')',
```

The unary minus production allows negative literals such as -5 and negated variable references such as -x. It is implemented as a right-recursive descent into factor and is desugared internally to `0 - factor`, so all existing type rules for arithmetic apply without modification.

2.4.3 Boolean Expressions

```
1 bool_expr  -> expr '==' expr
2           | expr '!=' expr
3           | BOOL_LITERAL
4           | expr (* fallback: type checker enforces Boolean *)
```

Boolean expressions are only parsed as conditions inside if and while statements. Comparison results are not storable in variables directly; the grammar does not include `bool_expr` in the factor rule. This is consistent with the assignment specification, which describes Boolean

expressions primarily as conditions.

If `bool_expr` parses an expression and finds no following `==` or `!=` operator, the expression is returned as-is and the type checker verifies that its inferred type is `Boolean`.

2.4.4 Statements

```
1 assignment -> IDENTIFIER '=' expr ';'
2 if_stmt    -> 'if' '(' bool_expr ')' block ('else' block)?
3 while_stmt -> 'while' '(' bool_expr ')' block
4 print_stmt -> 'print' '(' expr ')' ';'
5 return_stmt -> 'return' expr ';'
6 block      -> '{' statement* '}'
```

2.4.5 Function Definitions and Calls

```
1 func_def -> 'def' IDENTIFIER '(' params? ')' block
2 params   -> IDENTIFIER (',' IDENTIFIER)*
3 args     -> expr (',' expr)*
```

Function calls use value parameter passing. This means the argument values are evaluated before the call and copied into the function's local environment.

2.5 Operator Precedence and Associativity

Operator precedence is encoded structurally in the recursive-descent parser. Comparison is lowest, addition and subtraction are next, and multiplication and division have the highest precedence among binary operators. Arithmetic operators are left-associative.

Table 2.2. Operator Precedence (1 = lowest, 3 = highest) and Associativity

Precedence	Operators	Associativity
1 (lowest)	<code>==</code> , <code>!=</code>	non-associative (condition use only)
2	<code>+</code> , <code>-</code>	left-associative
3 (highest)	<code>*</code> , <code>/</code>	left-associative

CHAPTER 3

LEXICAL ANALYSIS AND SYMBOL TABLE

This chapter covers the responsibilities related to lexical and early semantic foundations of the language. It includes defining token structures and categories (such as keywords, operators, identifiers, and literals), implementing the lexer to transform source characters into tokens, and establishing the symbol table to manage variable and function storage along with their associated types. Together, these components ensure that raw source input is systematically organised into meaningful structures that can be used reliably by later stages of the compiler.

3.1 Token representation

The token system, written in `components/tokens.py`, establishes the vocabulary and structure of all lexical elements in the language. It specifies both the types of tokens that can exist and the structure each token must follow. This serves as a canonical contract between the lexer and the parser, ensuring consistency across all stages of the compiler.

Table 3.1 summarises the main categories, example lexemes, and typical `TokenType` values.

Table 3.1. Token categories and representative `TokenType` values

Category	Example lexemes	Representative <code>TokenType</code>
Literals	42, 3.14, true, "hello"	INT_LITERAL, FLOAT_LITERAL, BOOL_LITERAL, STRING_LITERAL
Identifiers	x, counter, add	IDENTIFIER
Keywords	if, else, while, def, return, print	keyword-specific token variants
Operators	+, -, *, /, =, ==, !=	PLUS, MINUS, STAR, SLASH, ASSIGN, EQUAL_EQUAL, BANG_EQUAL
Delimiters	Parentheses, braces, comma, and semicolon (see lexer delimiter tokens)	delimiter token variants
End marker	end of source input	EOF

Token types (from class `TokenType`) are defined using an Enum with `auto()` to automatically assign unique values. This avoids reliance on raw strings and ensures safer comparisons during parsing. The token categories include literals, identifiers, keywords, operators, delimiters, and a special end-of-file (EOF) token. The EOF token is appended after all input is processed,

allowing the parser to detect the end of input explicitly rather than encountering an unexpected termination.

Each token is represented as an immutable dataclass (`frozen=True`), meaning its fields cannot be modified after creation. Each token contains four key pieces of information: its kind (`token_type`), its lexeme (the exact substring from the source code), and its source position (`line`, `column`). This structured representation allows later stages to process the program in terms of syntax and errors without handling raw text directly. Immutability improves reliability by preventing accidental modification during later processing.

```
1 @dataclass(frozen=True)
2 class Token:
3     token_type: TokenType
4     lexeme: str
5     line: int
6     column: int
```

3.2 Lexer implementation

The lexer, implemented in `components/lexica.py`, is a hand-written scanner that converts raw source text into a sequence of tokens. Unlike generated lexers (for example SLY lexer mode, or Lex and Flex), this implementation uses explicit control flow (`if` and `while`) to process input.

The lexer maintains internal state including the source string, current position, start of the current token, line number, and column index. The column index is tracked per line and resets upon encountering a newline, while a separate `token_column` records the starting position of each token for accurate error reporting.

The main function, `tokenize`, iterates through the source code, repeatedly invoking `_scan_token` to identify the next token. Tokens are appended to a list, and an EOF token is added at the end.

Whitespace is skipped, while invalid input results in a `LexerError`.

```
1 def tokenize(self) -> list[Token]:
2     tokens: list[Token] = []
3     while not self._is_at_end():
4         self.start = self.current
5         self.token_column = self.column
6         token = self._scan_token()
7         if token is not None:
8             tokens.append(token)
9     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
10    return tokens
```

Token recognition follows a structured approach. Single-character tokens are mapped using a lookup table. Multi-character operators such as `==` and `!=` are handled using lookahead via `_match`. String literals are processed by `_string`, which reads characters until a closing quotation mark or the end of input is reached, with error handling for unterminated strings. Numeric literals are handled by `_number`, which distinguishes integers and floating-point values based on the presence of a decimal point followed by digits. Identifiers are processed greedily, consuming alphanumeric characters and underscores, and then checked against the `KEYWORDS` map to determine whether they are reserved words or user-defined names. Notably, `true` and `false` are also entries in this map, mapping to `TokenType.BOOL_LITERAL` rather than a keyword-specific type. This means boolean literals are recognised through the same keyword-fallback mechanism as reserved words, preventing them from being scanned as user-defined identifiers.

```
1 KEYWORDS: dict[str, TokenType] = {
2     "if": TokenType.IF,
3     "else": TokenType.ELSE,
4     "while": TokenType.WHILE,
5     # ...
6     "true": TokenType.BOOL_LITERAL,
7     "false": TokenType.BOOL_LITERAL,
8 }
```

```
1 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
2     "+": TokenType.PLUS,
3     "-": TokenType.MINUS,
4     "*": TokenType.STAR,
5     "/": TokenType.SLASH,
6     # ...
7 }
```

Helper functions such as `_advance`, `_peek`, and `_peek_next` manage character consumption and lookahead, while `_is_at_end` determines when input has been fully processed. Error messages include precise line and column information, aiding debugging and diagnostics.

Table 3.2 lists the main functions and data structures in `components/lexica.py` and their roles.

Table 3.2. Key functions and data in `lexica.py`

Function / data	Responsibility	Output / effect
<code>tokenize()</code>	Iterates through source, scans tokens, appends EOF	ordered token list
<code>_scan_token()</code>	Dispatches recognition based on character and lookahead	next token or skip
<code>_number()</code>	Parses numeric sequences, distinguishes int vs float	numeric literal token
<code>_string()</code>	Parses quoted text, checks termination	string token or error
<code>_identifier()</code>	Parses identifiers, resolves keyword vs user-defined	keyword or identifier token
KEYWORDS map	Maps reserved words to token types	token type selection
SINGLE_CHAR_TOKENS map	Maps single-character symbols	token type selection
<code>_advance()</code> , <code>_peek()</code> , <code>_match()</code>	Character consumption and lookahead	scanner state update

3.3 Symbol table and semantic structure

The symbol table, defined in `components/symbol_table.py`, operates at a higher level than the lexer. While the lexer processes raw text into tokens, the symbol table manages semantic information such as variable and function names, their types, and their scope.

The symbol table is implemented as a scoped structure, where each scope maintains its own dictionary of symbols and optionally references a parent scope. This enables proper handling of nested blocks and variable shadowing. Symbols are introduced through definition functions and retrieved through lookup operations, which search recursively through parent scopes if needed.

```

1 def lookup(self, name: str) -> Symbol:
2     symbol = self._symbols.get(name)
3     if symbol is not None:
4         return symbol
5     if self.parent is not None:
6         return self.parent.lookup(name)
7     raise SymbolTableError(f"Undefined symbol: {name}")

```

The type system is represented using a `LanguageType` enumeration, which includes fundamental types such as Integer, Float, Boolean, String, and Void. Symbols are modelled using an inheritance hierarchy: the base `Symbol` class stores a name and type, while subclasses such as `VariableSymbol` and `FunctionSymbol` include additional attributes. Variables track initialization state, while functions store parameter lists and return types. Table 3.3 summarises the main building blocks.

Table 3.3. Symbol table components and roles

Component	Stored information	Role in semantic analysis
Scoped symbol table	Symbols in current and parent scope	Supports nesting and shadowing via recursive lookup
LanguageType enum	Integer, Float, Boolean, String, Void	Defines the type system
Symbol (base)	Name and declared or inferred type	Base abstraction for entries
VariableSymbol	Variable metadata (including initialization state)	Tracks correct usage before assignment
FunctionSymbol	Parameters and return type	Validates calls and return consistency
lookup(name)	Recursive scope search	Resolves identifiers or reports undefined
SymbolTableError	Semantic error details	Reports duplicates, undefined symbols, invalid operations

Errors related to semantic violations are handled via `SymbolTableError`. These include duplicate declarations within the same scope, references to undefined symbols, and invalid operations such as attempting to mark a non-variable symbol as initialized.

3.4 Role of the symbol table in the pipeline

The symbol table is constructed during the type-checking phase rather than during lexical analysis. This guideline is followed in the implementation. The type checker initialises a global symbol table and populates it while traversing the AST, including defining variables and functions, managing nested scopes, and validating type correctness.

```

1 def run_pipeline(source_code: str) -> PipelineResult:
2     tokens = Lexer(source_code).tokenize()
3     tree = Parser(tokens).parse()
4     ast_text = ASTPrinter().print(tree)
5     checked_scope = TypeChecker().check(tree)
6     outputs: list[str] = []
7     Interpreter(output_callback=outputs.append).execute(tree)
8     return PipelineResult(
9         tokens=tokens,
10        ast_text=ast_text,
11        checked_scope=checked_scope,
12        outputs=outputs,
13    )

```

Once type checking is complete, the populated symbol table is returned as part of the pipeline result (`checked_scope`). The checker creates child scopes for blocks and functions during analysis, but the displayed `checked_scope` table represents the global scope entries returned

by the pipeline. It can then be displayed or used for further analysis.

This separation of concerns keeps lexical analysis focused on tokenization, while semantic meaning is enforced during type checking. For user-defined functions, definitions are first registered in the global scope with placeholder parameter and return types. These are inferred from the first valid call and refined through body checking; subsequent calls must match the inferred parameter types.

CHAPTER 4

RECURSIVE-DESCENT PARSING AND THE ABSTRACT SYNTAX TREE

This chapter presents the implementation of the recursive-descent parser and the associated abstract syntax tree (AST) model. Starting from lexer-produced tokens, the parser constructs a hierarchical Program AST that represents expressions, statements, and control-flow structure in a form suitable for downstream semantic analysis and execution. Implemented syntax coverage includes arithmetic expressions, assignments, if-then-else, while, function definitions, function calls, return, and built-in `print(...)` statements. Equality operators (`==`, `!=`) are parsed in condition contexts (if/while) through the parser’s boolean-expression path.

4.1 Abstract syntax tree representation

The AST is defined as a collection of Python dataclasses representing the structural elements of the programming language. It serves as an intermediate representation between parsing and later stages such as type checking and interpretation.

A key distinction exists between tokens and AST nodes. Tokens are flat and describe individual lexical elements such as operators or identifiers. In contrast, the AST is hierarchical. For example, while a token may represent the symbol `+`, a `BinaryOp` node represents an entire addition operation, including its left and right operands as subtrees.

All AST nodes inherit from a common base class, allowing them to store positional metadata (line, column). This ensures that errors detected in later stages can still reference precise locations in the original source code.

```
1 @dataclass
2 class ASTNode:
3     """Base class for every AST node. Carries source location for
4         diagnostics."""
5     line: int = field(default=0, kw_only=True)
6     column: int = field(default=0, kw_only=True)
```

4.1.1 Expressions

Expressions represent computations or values within the program. A `Literal` node stores values known at parse time, including integers, floating-point numbers, booleans, and strings. String literals are stored without surrounding quotation marks, as these are removed during parsing.

```
1 if tok.token_type == TokenType.STRING_LITERAL:
2     self._advance()
3     return Literal(value=tok.lexeme[1:-1], line=tok.line, column=tok.
        column)
```

An `Identifier` node represents the use of a variable name within an expression, indicating evaluation rather than declaration. A `BinaryOp` node models arithmetic and comparison operations such as addition, subtraction, multiplication, division, and equality checks, storing the operator along with references to left and right operands. A `FunctionCall` node represents function invocation, consisting of the function name and a list of argument expressions.

```
1 @dataclass
2 class BinaryOp(ASTNode):
3     """An arithmetic or comparison binary operation.
4
5     op is one of: '+', '-', '*', '/', '==', '!=',
6     Example:  a + b    x == 0
7     """
8     op: str
9     left: ASTNode
10    right: ASTNode
```

4.1.2 Statements

Statements represent actions or control flow within the program. An `Assign` node models variable assignment, pairing a variable name with a value expression, while `Print` and `Return` nodes each store a single expression and represent output and function return operations respectively. A `Block` node groups multiple statements within braces, representing a scoped execution sequence.

Control flow is handled via `If` and `While` nodes. The `If` node stores a condition, a then-block, and an optional else-block, while `While` stores a loop condition and body block. Function definitions are represented by `FunctionDef` nodes, which include the function name, a list of parameter names, and a body block. Parameter types are intentionally not assigned at parse time, as type inference is handled later by the type checker. Finally, `Program` serves as the

AST root, containing all top-level statements.

4.2 Parser implementation

The parser consumes the list of tokens generated by the lexer and produces a single Program AST node. It is implemented as a hand-written recursive-descent parser based on the project grammar. This avoids parser generators and instead relies on explicit functions for each grammar rule.

The parser maintains internal state consisting of the token list and a position index (`self._pos`) indicating the current token. Tokens are accessed via helper methods such as `_peek` and `_advance`. Parsing errors are reported using a custom `ParseError` exception with line and column information consistent with lexer diagnostics.

```
1 def parse(self) -> Program:
2     """Parse the full token stream and return the root Program node."""
3     line, col = self._peek().line, self._peek().column
4     statements: list[ASTNode] = []
5     while not self._is_at_end():
6         statements.append(self._statement())
7     return Program(statements=statements, line=line, column=col)
```

In addition to the main `parse()` entry point, the parser's statement dispatcher (`_statement`) implements a deterministic decision order over token patterns: function definition, conditional, loop, return, print, assignment, then expression statement fallback. This design is important because it prevents ambiguity between syntactically similar forms. In particular, assignments are recognized using one-token lookahead (`IDENTIFIER` followed by `=`), while identifier-led expressions that are not assignments are parsed as expression statements and must still end with a semicolon. This gives the parser full coverage of both declarative-style and expression-driven top-level statements without requiring a parser generator.

4.2.1 Statement parsing

The main parsing loop repeatedly processes statements until EOF, with the statement type determined via dispatch on the current token. Function definitions, conditionals, loops, returns, and prints each have dedicated parsing methods. Assignments are detected using one-token lookahead (`IDENTIFIER` followed by `ASSIGN`), preventing misclassification of function calls as assignments. If no specific statement pattern matches, the parser falls back to an expression statement and enforces a terminating semicolon.

```

1 # assignment: IDENTIFIER "=" expr ";"
2 if (
3     tok.token_type == TokenType.IDENTIFIER
4     and self._peek_next().token_type == TokenType.ASSIGN
5 ):
6     return self._assignment()
7 # fallback: bare expression statement expr ";"
8 node = self._expr()
9 self._expect(TokenType.SEMICOLON, "Expected ';' after expression")

```

4.2.2 Expression parsing and precedence

Expressions are parsed in layered precedence levels: `_expr` handles `+` and `-`, `_term` handles `*` and `/`, and `_factor` handles literals, identifiers and function calls, and parenthesized expressions. Boolean conditions for `if` and `while` are parsed by `_bool_expr`, which recognises `==` and `!=` after parsing the left-hand arithmetic expression. As a result, equality parsing is condition-focused in the current grammar implementation rather than a general infix operator available in every expression position. Function calls are recognised when an identifier is followed by `(`, with argument lists parsed as comma-separated expressions. Parenthesized expressions are supported to override precedence. Unary negation (`-x`, `-5`) is also handled at the `_factor` level: a leading `-` token causes the parser to recursively parse the operand and return `BinaryOp(0 - operand)`, so the existing type rules for subtraction cover negation without any additional inference code.

```

1 def _expr(self) -> ASTNode: # +, -
2     ...
3 def _term(self) -> ASTNode: # *, /
4     ...
5 def _factor(self) -> ASTNode: # unary -, literals, identifiers/calls,
6     grouped expr

```

Figure 4.1 shows the AST produced for the statement `z = x + y * 2;`. Because `_term` handles `*` before `_expr` handles `+`, the multiplication sub-tree is nested one level deeper, encoding the higher precedence of `*` over `+` without any explicit priority table at runtime.

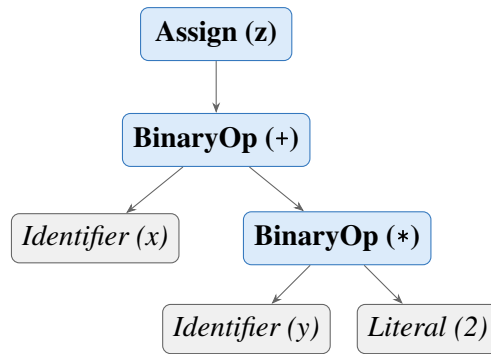


Figure 4.1. AST for $z = x + y * 2$; . Multiplication is a deeper sub-tree than addition, encoding higher precedence through tree structure rather than a runtime priority table.

Expression parsing is intentionally split into layered methods to enforce precedence and associativity in a transparent way. Parenthesized expressions are parsed explicitly and re-enter `_expr`, allowing precedence override where required by source syntax. Boolean conditions for `if` and `while` are parsed through `_bool_expr`, which accepts equality and inequality comparisons but can also return a non-comparison expression node; semantic enforcement that a condition is truly Boolean is deferred to the type checker.

4.2.3 *Helper functions and lookahead*

As described above, the parser uses `_check`, `_expect`, `_peek`, `_peek_next`, `_advance`, and `_is_at_end` for token navigation and validation. Lookahead is central for ambiguity resolution: assignment versus function call is resolved by inspecting the next token; comparison operators are recognised after parsing the left expression in `_bool_expr`. Parser errors follow a consistent diagnostic format: `[line X, col Y] ParseError: <message>`.

The parser also includes dedicated list parsers for formal parameters (`_params`) and call arguments (`_args`), both implemented as comma-separated sequences with optional emptiness. This ensures consistent handling of signatures such as `def f()` and `def f(a, b)` as well as invocations like `f()` and `f(x, y+1)`.

4.3 AST printing

For debugging and reporting, an AST printer converts the tree into a readable indented string, integrated into the pipeline output. The printer uses visitor-style dynamic dispatch, mapping node class names to methods such as `_visit_If` and `_visit_BinaryOp`, with output built as multi-line text where indentation reflects tree depth. Node formatting emphasises structure:

Program and Block include statement counts, BinaryOp explicitly shows left and right subtrees, and FunctionCall prints each argument by index. Literal includes both value and runtime Python type names for clarity.

4.4 Parser and type checker boundary

A deliberate architectural boundary is maintained between syntax construction and semantic validation. The parser is responsible for producing a structurally valid AST with accurate source locations, while type correctness and semantic constraints are checked later by the type checker. For example, `_bool_expr` may syntactically accept a non-Boolean expression as a condition node, but rejection of non-Boolean control-flow conditions is handled during semantic analysis. This separation keeps parser logic focused on grammar conformance and allows a clear responsibility split across team member contributions.

CHAPTER 5

TYPE INFERENCE ALGORITHM

5.1 Overview

Type inference in this project means that the programmer does not write explicit type declarations for variables or functions. Instead, the compiler infers types from assignments, literals, operations, function calls, and return statements. This keeps the language simple while still preserving static type safety.

The type checker runs after parsing. It receives the AST and validates the program before execution. Any inconsistent use of types is reported as a type error, and execution is stopped.

The interpreter stage does not repeat this validation: it walks the same AST with ordinary Python representations of values. Static type information remains in the symbol table produced by the checker; run-time values are not separately labelled with `LanguageType`, except that `print()` formats output by inspecting the value's Python type.

5.2 Type Environment

The type environment is represented by the symbol table. Conceptually, it is a mapping

$$\Gamma : Identifier \rightarrow Type$$

The environment grows as the checker visits the AST. On first assignment, a variable name is inserted with its inferred type. When a new block or a function call is entered, a child environment is created so that nested scopes can be checked independently while still accessing outer bindings when needed.

5.3 Type Inference Procedure

The type checker performs type inference by traversing the abstract syntax tree before interpretation. The algorithm is syntax-directed: each AST node is visited, the types of its child nodes are inferred first, and the current node is checked according to the language typing rules.

Algorithm TYPECHECK(*node*, Γ)

Input: *node* = current AST node; Γ = current type environment / symbol table

Output: inferred type of *node*, or `TypeError`

1. **If** *node* is a **Literal**: return the corresponding primitive type: `Integer`, `Float`, `Boolean`, or `String`.
2. **If** *node* is an **Identifier**: look up the identifier in Γ . If it is not found, report an undefined variable error. Otherwise, return its stored type.
3. **If** *node* is a **BinaryOp**:
 - Infer the type of the left operand and the right operand.
 - If the operator is `+`, `-`, `*`, or `/`: require both operands to be numeric; infer the result using Table 5.1.
 - If the operator is `==` or `!=`: require both operands to be arithmetic; return `Boolean`.
 - Otherwise, report a `TypeError`.
4. **If** *node* is an **Assignment**:
 - Infer the type of the right-hand-side expression.
 - If the variable is not yet in Γ : insert it with the inferred type.
 - Else: verify the new type matches the existing type; if not, report a `TypeError`.
5. **If** *node* is an **If** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the then-block in a child environment.
 - If an else-block exists, type-check it in a child environment.
6. **If** *node* is a **While** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the loop body in a child environment.
7. **If** *node* is a **FunctionDef**:
 - Register the function name in Γ .
 - Parameter types are initially unknown.
 - Return type is inferred from return statements in the body.
8. **If** *node* is a **FunctionCall**:

- Infer all argument types.
 - If this is the first valid call: store the argument types as the function’s parameter types.
 - Else: compare argument types with the stored parameter types.
 - Type-check the function body using a local environment.
 - Return the inferred function return type.
9. If *node* is a **Print** statement: infer the expression type; accept any of the four supported types.

5.4 Inference Rules

5.4.1 Literals

Literal nodes have direct types:

$$\begin{aligned}\Gamma \vdash 42 & : \text{Integer} \\ \Gamma \vdash 3.14 & : \text{Float} \\ \Gamma \vdash \text{true} & : \text{Boolean} \\ \Gamma \vdash \text{'hi'} & : \text{String}\end{aligned}$$

5.4.2 No Operator Overloading

The language does not support operator overloading. Every operator has a single, fixed meaning determined solely by the operator symbol. Specifically:

- The `+` operator is not overloaded for string concatenation. Applying `+` to a `String` operand is a type error.
- The comparison operators `==` and `!=` are not overloaded for string or boolean equality. Both operands must be arithmetic.
- No mixed Boolean or String arithmetic is permitted.

Because operators have unambiguous types, the compiler can always infer expression types without requiring explicit type declarations from the programmer.

5.4.3 Arithmetic Expressions

The arithmetic operators `+`, `-`, `*`, and `/` accept only numeric operands. If both operands are `Integer`, the result is `Integer`, except for division, which always produces `Float` regardless of the operand types. If at least one operand is `Float` for `+`, `-`, or `*`, the result is `Float` — this is numeric promotion, not operator overloading, because `+` retains a single numeric meaning throughout. Any non-numeric operand causes a type error.

Table 5.1. Arithmetic result types by operand combination

Left operand	Operator	Right operand	Result type
Integer	<code>+</code> , <code>-</code> , <code>*</code>	Integer	Integer
Integer	<code>/</code>	Integer	Float
Float	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Integer	Float
Integer	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Float	Float
Float	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Float	Float
Any	any	String or Boolean	<code>TypeError</code>

Unary negation

Unary minus (`-x`) is desugared by the parser into `0 - x`. The type checker therefore applies the standard binary subtraction rule: if `x` is `Integer`, the result is `Integer`; if `x` is `Float`, the result is `Float`.

5.4.4 Boolean Expressions

The comparison operators `==` and `!=` always produce `Boolean`. Both operators require two arithmetic (`Integer` or `Float`) operands; this constraint is a deliberate design choice that keeps comparison semantics unambiguous. Comparisons involving `Boolean` or `String` operands are rejected by the type checker. Although `Boolean` and `String` values exist in the language, equality and inequality are deliberately restricted to arithmetic expressions because the project specification defines basic `Boolean` expressions as equality and inequality between two arithmetic expressions.

5.4.5 Assignment Statement

Assignments define variable types on first use. If a variable does not yet exist in the type environment, the checker infers the type of the right-hand side and inserts the variable into the environment. If the variable already exists, the new value must have the same type.

If $x \notin \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma, x : T \vdash x = e$

If $x : T \in \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma \vdash x = e$

5.4.6 If-Then-Else

The condition of an `if` statement must be `Boolean`. The `then`-block and `else`-block are checked in child scopes so that local operations can be validated cleanly.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B_{\text{then}} \quad \Gamma'' \vdash B_{\text{else}} \implies \Gamma \vdash \text{if}(c) B_{\text{then}} \text{ else } B_{\text{else}}$$

5.4.7 While-Loop

The condition of a `while` loop must also be `Boolean`. The loop body is checked in a child scope and must not violate the types of variables already introduced in the surrounding environment.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B \implies \Gamma \vdash \text{while}(c) B$$

5.4.8 Function Definition and Call

Function parameter types are inferred from the first call to the function. After that first call, subsequent calls must use arguments of the same types. The return type is inferred from the `return` statements in the function body. If the function returns inconsistent types, the checker reports an error.

For functions that are defined but never called, the type checker performs a post-pass after all top-level statements have been processed. It infers parameter types by scanning the body for arithmetic and comparison usage, then checks the full body with those inferred types. This ensures that type errors inside never-called function bodies are still detected before execution.

Recursive calls are intentionally outside the scope of this implementation; supporting them would require additional fixed-point inference or explicit recursion handling that is orthogonal to the core type-inference algorithm.

5.5 Type Error Reporting

Type errors are reported with source positions so that the user can locate the problem quickly. The format used by the system is:

[line X, col Y] TypeError: message

Examples include assigning a String to an Integer variable, using a non-Boolean condition in an if statement, or supplying the wrong argument type to a function.

5.6 Example Type Inference Traces

Consider the following program fragment:

```
1 x = 10;
2 y = 20.5;
3 z = x + y;
4 print(z);
```

The checker infers:

- x: Integer
- y: Float
- x + y: Float
- z: Float

For a function example:

```
1 def add(a, b) {
2     return a + b;
3 }
4
5 result = add(2, 3.5);
```

The first function call infers a: Integer and b: Float. The expression a + b is therefore Float, so the function return type is inferred as Float. The variable result is also inferred as Float.

CHAPTER 6

SYSTEM ARCHITECTURE AND IMPLEMENTATION

This chapter describes how the implementation is organised: the end-to-end processing flow, repository layout, technology choices, and the main components that realise static analysis and execution (type checker, interpreter, and user interfaces). Detailed treatment of lexical analysis, the symbol table, and parsing appears in Chapters 3 and 4; Chapter 5 presents the type-inference rules that the type checker implements.

6.1 Compiler pipeline

The system follows a four-stage pipeline. Lexical and syntactic processing are kept separate from semantic analysis and execution, which simplifies testing and localises errors to the appropriate stage.

The overall flow of data through the system is shown in Figure 6.1.

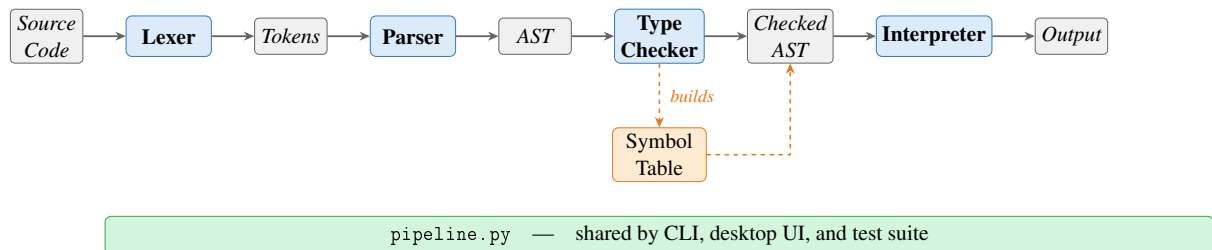


Figure 6.1. Compiler pipeline overview.

Source characters are scanned by the lexer into a token stream. The parser consumes tokens and builds an AST that reflects program structure (expressions, statements, and control flow). The type checker then walks the AST, populates and uses the symbol table, infers types, and validates scope and type rules. Only programs that pass the type checker proceed to interpretation. The interpreter evaluates the checked tree using ordinary Python values; it does not re-run the type rules, since correctness at runtime is guaranteed by the prior static pass. All four stages are orchestrated through `pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite.

6.1.1 Entry points and shared pipeline

All stages are orchestrated by `components/pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite. From the project root, `python3 main.py` runs a script (or built-in sample) and prints the stages in order: tokens, AST text, type table, and execution output. `python3 ui.py` runs the Tkinter workbench: the same pipeline is used internally, with each stage shown in a separate tab so the user can inspect results without altering the underlying flow.

6.2 Project structure

The main source files and their responsibilities are summarised in Table 6.1.

Table 6.1. Main source files

File	Responsibility
<code>main.py</code>	Command-line runner for sample code or a script file
<code>ui.py</code>	Desktop UI for editing and running a script
<code>components/tokens.py</code>	Token classes and token categories
<code>components/lexica.py</code>	Lexical analysis
<code>components/ast_nodes.py</code>	AST node definitions
<code>components/parser.py</code>	Recursive-descent parser
<code>components/ast_printer.py</code>	AST renderer for debugging and reporting
<code>components/symbol_table.py</code>	Scoped symbol table and semantic types
<code>components/type_checker.py</code>	Static type checker and inference
<code>components/interpreter.py</code>	Tree-walking interpreter
<code>components/pipeline.py</code>	Shared pipeline for CLI, UI, and tests

6.3 Technology stack

The project uses Python and standard-library tools only (Table 6.2).

Table 6.2. Technology stack

Tool	Purpose
Python 3.10+	Core implementation language
<code>dataclasses</code>	AST and runtime helper structures
<code>tkinter</code>	Desktop graphical interface
<code>unittest</code>	Automated testing

6.4 Type checker

The `TypeChecker` class implements the inference and checking rules described in Chapter 5. It traverses the AST before execution, uses the symbol table to record variable and function information, infers types on first assignment, validates arithmetic and comparison expressions, enforces Boolean conditions for control flow, infers function parameter types from the first call (or from body usage for uncalled functions), and infers function return types from `return` statements. Programs that fail these rules are rejected before interpretation.

6.4.1 Scope of the implemented type system

In scope terms, this component turns a syntactically valid AST into a semantically valid program:

- static typing rules for arithmetic, comparison, assignment, control flow, and functions,
- type checking for expressions and statements,
- variable type inference from first assignment,
- function parameter inference from the first call (or from body usage for uncalled functions),
- function return type inference from `return` statements,
- source-positioned type errors when a program violates a typing rule.

6.5 Interpreter

The interpreter is a tree-walking evaluator over the checked AST. It maintains nested runtime environments for variables and function calls. Assignments write through the current or nearest enclosing scope; `if` selects a branch; `while` repeats until its condition becomes false; function calls bind arguments by value and return through an internal return mechanism.

The built-in `print()` function outputs both value and runtime type. All four types are supported:

```
1 print(42);           (* 42 : Integer      *)
2 print(3.14);         (* 3.14 : Float      *)
3 print(true);         (* true : Boolean    *)
4 print("plc");        (* "plc" : String   *)
```

6.5.1 Scope of the implemented runtime

The runtime stage provides observable behaviour and integrates with the shared pipeline:

- assignment execution,
- if execution,
- while execution,
- function execution with value parameter passing,
- `print()` execution with immediate output,
- execution behind the same `run_pipeline` path used by the CLI, UI, and tests.

6.6 Graphical user interface

The desktop workbench (`ui.py`) is implemented with Python's `tkinter` standard library and requires no additional dependencies. The window is titled *125970-126690-126687-Ass2-PLC* and opens at 1200×760 pixels in a horizontal split layout.

Toolbar. A top toolbar provides three buttons — *Open Script*, *Run*, and *Clear Output* — and a right-aligned status label that reports *Ready*, a loaded filename, or *Run failed*.

Left panel (source editor). A monospaced `Text` widget with both horizontal and vertical scrollbars allows the user to enter or paste arbitrary source code. The *Open Script* button populates the editor from a file. The editor is pre-loaded with the built-in sample program on first launch.

Right panel (output tabs). A `Notebook` widget presents four tabs, each backed by a scrollable read-only `Text` widget with both horizontal and vertical scrollbars:

- *Execution Output* — the `print()` values produced by stage 4, formatted as `value : Type`;
- *Tokens* — the token stream from stage 1, one token per line with type, lexeme, and source position;
- *AST* — the indented tree text from stage 2;

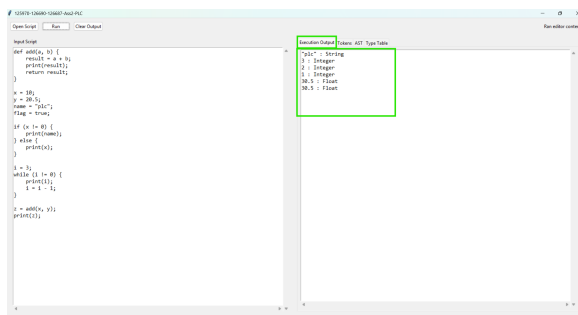
- *Type Table* — the symbol table from stage 3 showing each name, kind, inferred type, initialisation state, and parameter list.

Error handling. If any pipeline stage raises an exception, the error message is written to the *Execution Output* tab as ERROR: <message>. No modal dialogs are used, so the user can edit and rerun without dismissing a popup.

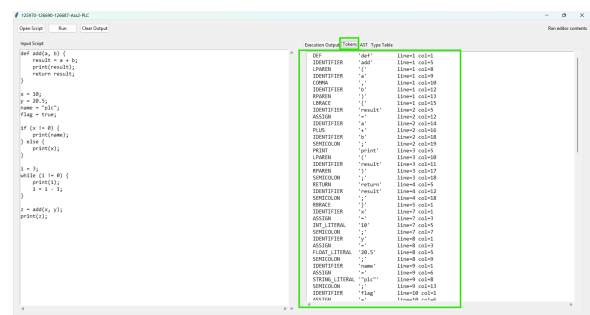
Keyboard shortcuts. F5 and Ctrl+Enter are both bound to the *Run* action, allowing rapid edit-run cycles without reaching for the mouse.

6.6.1 Screenshots

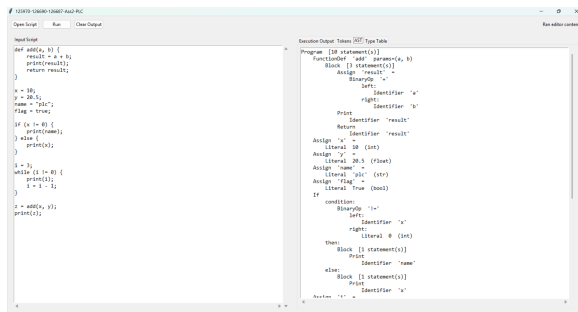
Figures 6.2 and 6.3 show the desktop workbench and CLI output.



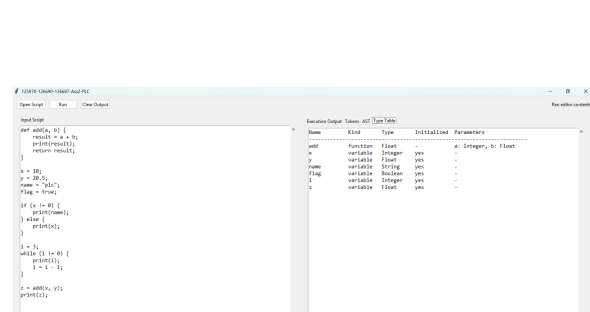
(a) Execution Output tab.



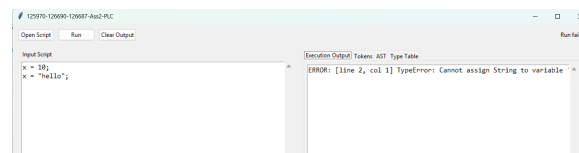
(b) Tokens tab (Stage 1).



(c) AST tab (Stage 2).



(d) Type Table tab (Stage 3).



(e) Inline type-error display.

Figure 6.2. Desktop workbench — output tabs and error display.

6.7 Command-line interface

The CLI runner (`main.py`) accepts an optional script path; without one it runs the built-in sample and prints four labelled stages to `stdout`.

```
(.venv) PS C:\Users\Windows\Desktop\125970-126690-126687-Assignment2-PLC> python main.py
=====
STAGE 1 - TOKENS
=====
DEF          'def'          line=2 col=1
IDENTIFIER   'add'          line=2 col=5
LPAREN      '('            line=2 col=8
IDENTIFIER   'a'            line=2 col=9
COMMA       ','            line=2 col=10
IDENTIFIER   'b'            line=2 col=12
RPAREN      ')'            line=2 col=13
LBRACE      '{'            line=2 col=15
IDENTIFIER   'result'       line=3 col=5
ASSIGN      '='            line=3 col=12
IDENTIFIER   'a'            line=3 col=14
PLUS        '+'            line=3 col=16
IDENTIFIER   'b'            line=3 col=18
SEMICOLON   ';'            line=3 col=19
PRINT       'print'        line=4 col=5
LPAREN      '('            line=4 col=10
IDENTIFIER   'result'       line=4 col=11
RPAREN      ')'            line=4 col=17
SEMICOLON   ';'            line=4 col=18
RETURN      'return'       line=5 col=5
IDENTIFIER   'result'       line=5 col=12
SEMICOLON   ';'            line=5 col=18
RBRACE      '}'            line=6 col=1
IDENTIFIER   'x'            line=8 col=1
ASSIGN      '='            line=8 col=3
INT_LITERAL '10'           line=8 col=5
SEMICOLON   ';'            line=8 col=7
IDENTIFIER   'y'            line=9 col=1
ASSIGN      '='            line=9 col=3
FLOAT_LITERAL '20.5'        line=9 col=5
SEMICOLON   ';'            line=9 col=9
IDENTIFIER   'name'         line=10 col=1
ASSIGN      '='            line=10 col=6
STRING_LITERAL 'plc'        line=10 col=8
SEMICOLON   ';'            line=10 col=13
=====
```

(a) Stage 1: token stream.

```

Assign 'i' =
  BinaryOp '-'
    left:
      Identifier 'i'
    right:
      Literal 1 (int)

Assign 'z' =
  FunctionCall 'add' [2 arg(s)]
    arg[0]:
      Identifier 'x'
    arg[1]:
      Identifier 'y'

Print
  Identifier 'z'
=====
STAGE 3 - TYPE CHECK
=====
Name      Kind      Type      Initialized Parameters
-----
add       function  Float     -           a: Integer, b: Float
x         variable  Integer   yes          -
y         variable  Float     yes          -
name      variable  String    yes          -
flag      variable  Boolean   yes          -
i         variable  Integer   yes          -
z         variable  Float     yes          -
=====
STAGE 4 - EXECUTION
=====
"plc" : String
3 : Integer
2 : Integer
1 : Integer
30.5 : Float
30.5 : Float
(.venv) PS C:\Users\Windows\Desktop\125970-126690-126687-Assignment2-PLC>
=====
```

(b) Stage 3 type table and Stage 4 output.

Figure 6.3. CLI run of python main.py on the built-in sample.

CHAPTER 7

TESTING AND VERIFICATION

7.1 Manual Testing

Prior to the construction of the automated suite, each language feature was exercised manually by running the full pipeline and inspecting stage outputs. Table 7.1 summarises the categories covered.

Table 7.1. Representative manually tested categories

Category	Program Behaviour	Expected Result
Arithmetic	Evaluate mixed numeric expressions	Correct value and inferred type
If-then-else	Choose one branch from a Boolean condition	Only matching branch runs
While-loop	Print during repeated execution	One output line per iteration
Functions	Infer parameters and return type	Valid call and correct return value
Type mismatch	Reassign incompatible type	Type error before runtime

7.1.1 *Lexer and Symbol Table*

The lexer was verified manually by running the full pipeline and inspecting Stage 1 token output for programs that exercise every token category: integer and float literals, Boolean literals, string literals, all six keywords, two-character comparison operators (`==`, `!=`), single-character operators, and delimiters. Identifier scanning with keyword fallback was confirmed by checking that `true`, `false`, and reserved words are emitted with their keyword token types while non-keyword names produce `IDENTIFIER` tokens. For example, the source fragment `x = 10;` produces the token sequence:

```

1 IDENTIFIER  'x'      line=1 col=1
2 ASSIGN      '='      line=1 col=3
3 INT_LITERAL '10'     line=1 col=5
4 SEMICOLON   ';'      line=1 col=7

```

Error paths were also checked: supplying a bare `!` character raises a `LexerError` with a source

position and a message suggesting !=, and an unterminated string literal raises a `LexerError` citing the opening line.

The symbol table was verified through the type checker and the Stage 3 output. Duplicate variable definitions were confirmed to raise a `SymbolTableError` before execution. Scope isolation was verified with nested blocks: a variable defined inside an `if` or `while` body is not visible in the outer scope after the block ends. The `format_table()` output was checked against the expected column layout (name, kind, type, initialisation status, parameters) for both variable and function symbols.

7.1.2 *Parser and AST*

The parser was verified manually by running the full pipeline and inspecting Stage 2 AST output. The following cases were checked:

- arithmetic expressions with mixed precedence (e.g. `x + y * 2`) produce a tree where multiplication is a deeper sub-tree than addition, confirming the `_expr/_term/_factor` precedence encoding,
- assignment versus expression statement dispatch: `x = 5`; produces an `Assign` node while `add(1,2)`; produces a `FunctionCall` node at the statement level, confirming the IDENTIFIER+ASSIGN lookahead,
- `if` with and without an `else` clause: the resulting `If` node has a non-null `else_block` only when the clause is present,
- function definitions store only parameter names in the `FunctionDef` node; the AST printer output was inspected to confirm no type annotations are present at this stage,
- string literals in `Literal` nodes have their surrounding double-quotes stripped; `print("hello")` prints `"hello"` : `String` at runtime (the interpreter re-adds the quotes when formatting string output),
- parse errors: missing semicolons, unmatched parentheses, and unexpected tokens all produce `[line X, col Y] ParseError:` messages with correct source positions.

7.1.3 Arithmetic Expressions

Arithmetic expressions were tested with integer-only and mixed integer–float programs to verify operator precedence and numeric promotion. The expression `x + y * 2` confirmed that multiplication binds more tightly than addition, and mixed-type operands consistently produced `Float` results in accordance with the promotion rules defined in Chapter 5.

7.1.4 Boolean Expressions

Boolean expressions were tested inside `if` and `while` conditions. The type checker correctly enforced that both operands of `==` and `!=` must be arithmetic; supplying a `Boolean` or `String` operand produced a source-positioned `TypeError` before any execution.

7.1.5 Assignment Statements

Assignments were tested for both first-use type inference and reassignment consistency. Assigning 5 to a variable and subsequently assigning "hello" to the same variable correctly produced a type error, confirming that the inferred type is fixed after first assignment.

7.1.6 If-Then-Else

Conditional programs were constructed to verify that exactly one branch executes for a given condition and that neither branch is entered when the condition evaluates to the opposite truth value. Non-Boolean conditions were also supplied and confirmed to be rejected by the type checker before execution.

7.1.7 While-Loop

While-loops were tested with an integer countdown from three to zero. The results confirmed that the loop body executes once per iteration and terminates when the condition becomes false, with `print()` producing one output line per pass rather than accumulating output until loop exit.

7.1.8 Functions

Function tests covered value parameter passing, local scope execution, and return value propagation. The type checker correctly inferred parameter types from the first call site and enforced that signature for all subsequent calls to the same function.

7.1.9 *Print*

The built-in `print()` function was tested with all supported runtime types. The output includes the printed value together with its type, such as `29.5 : Float` and `"plc" : String`.

7.1.10 *Unary Minus*

The unary minus operator was verified for integer literals (`x = -5`), float literals (`y = -3.14`), negated variable references (`y = -x`), and negation inside a larger expression (`x = 3 + -2`). In each case the type checker assigned the correct type and the interpreter produced the expected value.

7.2 Type Error Detection

The checker was verified with invalid programs involving incompatible assignment, invalid arithmetic operands, wrong function arguments, and non-Boolean conditions. In each case the system stopped before execution and reported a source-positioned type error. Representative error messages are shown below.

```
1 # Assignment type mismatch
2 [line 2, col 1] TypeError: Cannot assign String to variable 'x' of type
   Integer
3
4 # Non-Boolean if condition
5 [line 1, col 4] TypeError: If condition must have type Boolean
6
7 # Wrong argument type for function
8 [line 5, col 9] TypeError: Argument 1 of 'add' must be Integer, got
   String
```

7.3 Automated Test Suite

The automated regression suite is located in `tests/test_pipeline.py` and uses Python's built-in `unittest` framework. The tests are organised into five classes, each targeting a distinct compiler stage.

Table 7.2. Automated test coverage by class

Test class	Stage covered	Tests
<code>LexerTests</code>	Tokenisation (<code>lexica.py</code>)	10
<code>SymbolTableTests</code>	Scoped symbol table (<code>symbol_table.py</code>)	7
<code>ParserTests</code>	Parsing and precedence (<code>parser.py</code>)	5
<code>TypeCheckerTests</code>	Static type inference (<code>type_checker.py</code>)	15
<code>IntegrationTests</code>	Full pipeline execution	17
Total		54

LexerTests verify that each token category (integer, float, Boolean, string, keywords, identifiers, two-character operators) is produced correctly, that source line numbers are tracked across newlines, and that illegal characters and unterminated strings raise `LexerError`.

SymbolTableTests exercise the symbol table directly, without going through the full pipeline. They cover variable definition and lookup, duplicate definition raising `SymbolTableError` for both variables and functions, parent-scope lookup through `child_scope()`, undefined-symbol lookup error, and the `format_table()` output for both variable and function symbol rows.

TypeCheckerTests verify static type inference and error detection. The 15 tests cover integer and float arithmetic, division producing `Float`, numeric promotion, string and boolean type inference, assignment type mismatch, non-Boolean conditions, undefined variables, wrong argument count and type, and two tests for uncalled-function body checking: one confirms that a type error inside a never-called function body is detected, and another confirms that a valid uncalled function raises no error.

IntegrationTests run programs through all four pipeline stages and verify the final output. Cases include unary negation on all numeric types, both branches of `if/else`, while-loop countdown, `print()` with every type, value-parameter isolation, nested function calls, variable reassignment, and scope isolation (variables declared inside an `if` block are not visible outside it).

The tests can be run with:

```
python3 -m unittest discover -s tests -v
```

All 54 tests pass.

7.4 Error Handling

The language system reports errors at three main stages:

- lexical errors for malformed characters or unterminated strings,
- parse errors for invalid syntax such as missing semicolons,
- type errors for programs that violate static typing rules.

Runtime errors are also reported with source positions whenever execution reaches an invalid state such as an undefined variable reference.

CHAPTER 8

CONCLUSION

This project delivered a complete, statically-typed interpreted language supporting Integer, Float, Boolean, and String types, with arithmetic, comparisons, assignments, if-then-else, while-loops, functions with value parameter passing, and a built-in `print()` function.

The system follows a four-stage pipeline: lexer, parser, type checker, and interpreter. Variable and function types are inferred without explicit declarations; programs that violate type rules are rejected before execution. For uncalled functions, the type checker performs a post-pass to infer parameter types from body usage, ensuring type errors are always caught.

The 54 automated tests across five classes — lexical analysis, symbol table, parsing, type checking, and full pipeline — confirm correct behaviour for all language features. Both a command-line runner and a desktop UI are provided.

The implementation demonstrates all core compiler-construction concepts covered in the course — lexical analysis, parsing, static type inference, and tree-walking interpretation — in a complete, tested, and working system. Future work could extend the language with recursive functions, richer data structures, and code generation.

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APPENDIX A: SOURCE CODE

The complete source code is available on GitHub:

<https://github.com/The-Token-Trio/125970-126690-126687-Assignment2-PLC>

The project can be run from the command line or the desktop UI:

```
1 python3 main.py
2 python3 ui.py
3 python3 -m unittest discover -s tests -v
```

No external Python dependencies are required.

APPENDIX B: TEAM MEMBERS AND CONTRIBUTIONS

Name	Student ID	Contributions
Aye Khin Khin Hpone (Yolanda Lim)	st125970	Static typing rules; type checking; assignment, if, while, function, and <code>print()</code> execution; unary minus inference; pipeline integration (<code>pipeline.py</code>); automated test suite (54 tests)
Applegate T. Tun Oo	st126690	Lexer implementation (scanning, tokenisation, line/column tracking, error reporting); token definitions; keywords, operators, identifiers, and literals; symbol table (variable/function/type storage)
Win Htut Naing	st126687	Arithmetic and boolean expressions; assignment, if-then-else, while-loop, function definition/call, and <code>print()</code> parsing; unary minus parsing

A.1 Entry Points

```
1 from __future__ import annotations
2
3 import argparse
4 from pathlib import Path
5
6 from components.pipeline import format_stage_output, run_pipeline
7
8
```

```

9  DEFAULT_SOURCE = """
10 def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 x = 10;
17 y = 20.5;
18 name = "plc";
19 flag = true;
20
21 if (x != 0) {
22     print(name);
23 } else {
24     print(x);
25 }
26
27 i = 3;
28 while (i != 0) {
29     print(i);
30     i = i - 1;
31 }
32
33 z = add(x, y);
34 print(z);
35 """
36
37
38 def run_source(source_code: str) -> None:
39     result = run_pipeline(source_code)
40     print(format_stage_output(result))
41
42
43 def _parse_args() -> argparse.Namespace:
44     parser = argparse.ArgumentParser(description="Run the programming
45         language pipeline.")
46     parser.add_argument("script", nargs="?", help="Optional path to a
47         source file to run")
48     return parser.parse_args()
49
50 def main() -> None:
51     args = _parse_args()
52     if args.script:
53         source_code = Path(args.script).read_text(encoding="utf-8")
54     else:
55         source_code = DEFAULT_SOURCE
56     run_source(source_code)
57
58 if __name__ == "__main__":
59     main()

```

Listing 8.1: main.py — command-line entry point

```
1 from __future__ import annotations
2
3 import tkinter as tk
4 from pathlib import Path
5 from tkinter import filedialog, ttk
6
7 from components.pipeline import format_tokens, run_pipeline
8
9
10 DEFAULT_SOURCE = """def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 x = 10;
17 y = 20.5;
18 name = \"plc\";
19 flag = true;
20
21 if (x != 0) {
22     print(name);
23 } else {
24     print(x);
25 }
26
27 i = 3;
28 while (i != 0) {
29     print(i);
30     i = i - 1;
31 }
32
33 z = add(x, y);
34 print(z);
35 """
36
37
38 class LanguageWorkbench:
39     def __init__(self) -> None:
40         self.root = tk.Tk()
41         self.root.title("125970-126690-126687-Ass2-PLC")
42         self.root.geometry("1200x760")
43         self.current_file: Path | None = None
44
45         self._build_layout()
46         self.source_text.insert("1.0", DEFAULT_SOURCE)
47         self.root.bind("<F5>", lambda _e: self._run_source())
48         self.root.bind("<Control-Return>", lambda _e: self._run_source
49             ())
```

```

50 def _build_layout(self) -> None:
51     toolbar = ttk.Frame(self.root, padding=10)
52     toolbar.pack(fill=tk.X)
53
54     ttk.Button(toolbar, text="Open Script", command=self.
55         _open_script).pack(side=tk.LEFT)
56     ttk.Button(toolbar, text="Run", command=self._run_source).pack(
57         side=tk.LEFT, padx=(8, 0))
58     ttk.Button(toolbar, text="Clear Output", command=self.
59         _clear_output).pack(side=tk.LEFT, padx=(8, 0))
60
61     self.status_var = tk.StringVar(value="Ready")
62     ttk.Label(toolbar, textvariable=self.status_var).pack(side=tk.
63         RIGHT)
64
65     body = ttk.Panedwindow(self.root, orient=tk.HORIZONTAL)
66     body.pack(fill=tk.BOTH, expand=True, padx=10, pady=(0, 10))
67
68     left_panel = ttk.Frame(body, padding=8)
69     right_panel = ttk.Frame(body, padding=8)
70     body.add(left_panel, weight=1)
71     body.add(right_panel, weight=1)
72
73     ttk.Label(left_panel, text="Input Script").pack(anchor=tk.W)
74     src_frame = ttk.Frame(left_panel)
75     src_frame.pack(fill=tk.BOTH, expand=True, pady=(6, 0))
76     self.source_text = tk.Text(src_frame, wrap=tk.NONE, font=(
77         "Consolas", 11))
78     src_vsb = ttk.Scrollbar(src_frame, orient=tk.VERTICAL, command=
79         self.source_text.yview)
80     src_hsb = ttk.Scrollbar(src_frame, orient=tk.HORIZONTAL,
81         command=self.source_text.xview)
82     self.source_text.configure(yscrollcommand=src_vsb.set,
83         xscrollcommand=src_hsb.set)
84     src_vsb.pack(side=tk.RIGHT, fill=tk.Y)
85     src_hsb.pack(side=tk.BOTTOM, fill=tk.X)
86     self.source_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
87
88     notebook = ttk.Notebook(right_panel)
89     notebook.pack(fill=tk.BOTH, expand=True)
90
91     self.output_text = self._make_output_tab(notebook, "Execution
    Output")
92     self.tokens_text = self._make_output_tab(notebook, "Tokens")
93     self.ast_text = self._make_output_tab(notebook, "AST")
94     self.types_text = self._make_output_tab(notebook, "Type Table")
95
96 def _make_output_tab(self, notebook: ttk.Notebook, title: str) ->
    tk.Text:
97     frame = ttk.Frame(notebook, padding=8)
98     notebook.add(frame, text=title)
99     text = tk.Text(frame, wrap=tk.NONE, font=("Consolas", 11),
100         state=tk.DISABLED)

```

```

92     vsb = ttk.Scrollbar(frame, orient=tk.VERTICAL, command=text.
93         yview)
94     hsb = ttk.Scrollbar(frame, orient=tk.HORIZONTAL, command=text.
95         xview)
96     text.configure(yscrollcommand=vsb.set, xscrollcommand=hsb.set)
97     vsb.pack(side=tk.RIGHT, fill=tk.Y)
98     hsb.pack(side=tk.BOTTOM, fill=tk.X)
99     text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
100     return text
101
102 def _open_script(self) -> None:
103     file_path = filedialog.askopenfilename(
104         title="Open Script File",
105         filetypes=[("Text Files", "*.txt *.pl *.src"), ("All Files",
106             ".*")],
107     )
108     if not file_path:
109         return
110
111     path = Path(file_path)
112     self.current_file = path
113     self.source_text.delete("1.0", tk.END)
114     self.source_text.insert("1.0", path.read_text(encoding="utf-8"))
115
116     self.status_var.set(f"Loaded {path.name}")
117
118 def _run_source(self) -> None:
119     source = self.source_text.get("1.0", tk.END)
120     try:
121         result = run_pipeline(source)
122     except Exception as error:
123         self._write_text(self.output_text, f"ERROR: {error}")
124         self.status_var.set("Run failed")
125         return
126
127     self._write_text(self.output_text, "\n".join(result.outputs) if
128         result.outputs else "(no output)")
129     self._write_text(self.tokens_text, format_tokens(result.tokens))
130
131     self._write_text(self.ast_text, result.ast_text)
132     self._write_text(self.types_text, result.checked_scope.
133         format_table())
134
135     current_label = self.current_file.name if self.current_file is
136         not None else "editor contents"
137     self.status_var.set(f"Ran {current_label}")
138
139 def _clear_output(self) -> None:
140     self._write_text(self.output_text, "")
141     self._write_text(self.tokens_text, "")
142     self._write_text(self.ast_text, "")
143     self._write_text(self.types_text, "")
144     self.status_var.set("Output cleared")

```

```

137
138     @staticmethod
139     def _write_text(widget: tk.Text, content: str) -> None:
140         widget.config(state=tk.NORMAL)
141         widget.delete("1.0", tk.END)
142         widget.insert("1.0", content)
143         widget.config(state=tk.DISABLED)
144
145     def run(self) -> None:
146         self.root.mainloop()
147
148
149 if __name__ == "__main__":
150     LanguageWorkbench().run()

```

Listing 8.2: ui.py — Tkinter desktop workbench

A.2 Core Components

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from enum import Enum, auto
5
6
7 class TokenType(Enum):
8     # Literals
9     INT_LITERAL = auto()
10    FLOAT_LITERAL = auto()
11    BOOL_LITERAL = auto()
12    STRING_LITERAL = auto()
13
14    # Identifier
15    IDENTIFIER = auto()
16
17    # Keywords
18    IF = auto()
19    ELSE = auto()
20    WHILE = auto()
21    DEF = auto()
22    RETURN = auto()
23    PRINT = auto()
24
25    # Operators
26    PLUS = auto()
27    MINUS = auto()
28    STAR = auto()
29    SLASH = auto()
30    EQUAL_EQUAL = auto()
31    BANG_EQUAL = auto()
32    ASSIGN = auto()
33

```

```

34     # Delimiters
35     LPAREN = auto()
36     RPAREN = auto()
37     LBRACE = auto()
38     RBRACE = auto()
39     COMMA = auto()
40     SEMICOLON = auto()
41
42     EOF = auto()
43
44
45 @dataclass(frozen=True)
46 class Token:
47     token_type: TokenType
48     lexeme: str
49     line: int
50     column: int

```

Listing 8.3: components/tokens.py — token type enumeration and Token dataclass

```

1  from __future__ import annotations
2
3  from typing import Iterator
4
5  from components.tokens import Token, TokenType
6
7
8  KEYWORDS: dict[str, TokenType] = {
9      "if": TokenType.IF,
10     "else": TokenType.ELSE,
11     "while": TokenType.WHILE,
12     "def": TokenType.DEF,
13     "return": TokenType.RETURN,
14     "print": TokenType.PRINT,
15     "true": TokenType.BOOL_LITERAL,
16     "false": TokenType.BOOL_LITERAL,
17 }
18
19
20 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
21     "+": TokenType.PLUS,
22     "-": TokenType.MINUS,
23     "*": TokenType.STAR,
24     "/": TokenType.SLASH,
25     "=": TokenType.ASSIGN,
26     "(": TokenType.LPAREN,
27     ")": TokenType.RPAREN,
28     "{": TokenType.LBRACE,
29     "}": TokenType.RBRACE,
30     ",": TokenType.COMMA,
31     ";": TokenType.SEMICOLON,
32 }
33

```

```

34
35 class LexerError(Exception):
36     pass
37
38
39 class Lexer:
40     def __init__(self, source: str) -> None:
41         self.source = source
42         self.start = 0
43         self.current = 0
44         self.line = 1
45         self.column = 1
46         self.token_column = 1
47
48     def tokenize(self) -> list[Token]:
49         tokens: list[Token] = []
50         while not self._is_at_end():
51             self.start = self.current
52             self.token_column = self.column
53             token = self._scan_token()
54             if token is not None:
55                 tokens.append(token)
56         tokens.append(Token(TokenType.EOF, "", self.line, self.column))
57         return tokens
58
59     def _scan_token(self) -> Token | None:
60         char = self._advance()
61
62         # Whitespace handling
63         if char in {" ", "\t", "\r"}:
64             return None
65         if char == "\n":
66             self.line += 1
67             self.column = 1
68             return None
69
70         # Two-char operators
71         if char == "=":
72             if self._match("="):
73                 return self._make_token(TokenType.EQUAL_EQUAL)
74             return self._make_token(TokenType.ASSIGN)
75         if char == "!":
76             if self._match("="):
77                 return self._make_token(TokenType.BANG_EQUAL)
78             raise LexerError(self._error_message("Unexpected '!' . Did
              you mean '!='?"))
79
80         # Single-char operators/delimiters
81         single_char_token = SINGLE_CHAR_TOKENS.get(char)
82         if single_char_token is not None:
83             return self._make_token(single_char_token)
84
85         if char == "'":

```

```

86         return self._string()
87
88     if char.isdigit():
89         return self._number()
90
91     if self._is_identifier_start(char):
92         return self._identifier()
93
94     raise LexerError(self._error_message(f"Unexpected character: {
95         char!r}"))
96
97 def _string(self) -> Token:
98     while self._peek() != '"' and not self._is_at_end():
99         if self._peek() == "\n":
100             self.line += 1
101             self.column = 1
102             self._advance()
103
104     if self._is_at_end():
105         raise LexerError(self._error_message("Unterminated string
106             literal"))
107
108     # Closing quote
109     self._advance()
110     return self._make_token(TokenType.STRING_LITERAL)
111
112 def _number(self) -> Token:
113     while self._peek().isdigit():
114         self._advance()
115
116     is_float = False
117     if self._peek() == "." and self._peek_next().isdigit():
118         is_float = True
119         self._advance()
120         while self._peek().isdigit():
121             self._advance()
122
123     if is_float:
124         return self._make_token(TokenType.FLOAT_LITERAL)
125     return self._make_token(TokenType.INT_LITERAL)
126
127 def _identifier(self) -> Token:
128     while self._is_identifier_part(self._peek()):
129         self._advance()
130
131     lexeme = self._current_lexeme()
132     token_type = KEYWORDS.get(lexeme, TokenType.IDENTIFIER)
133     return self._make_token(token_type)
134
135 def _current_lexeme(self) -> str:
136     return self.source[self.start : self.current]
137
138 def _make_token(self, token_type: TokenType) -> Token:

```

```

137         return Token(token_type, self._current_lexeme(), self.line,
138                        self.token_column)
139
140     def _advance(self) -> str:
141         char = self.source[self.current]
142         self.current += 1
143         self.column += 1
144         return char
145
146     def _match(self, expected: str) -> bool:
147         if self._is_at_end():
148             return False
149         if self.source[self.current] != expected:
150             return False
151
152         self.current += 1
153         self.column += 1
154         return True
155
156     def _peek(self) -> str:
157         if self._is_at_end():
158             return "\0"
159         return self.source[self.current]
160
161     def _peek_next(self) -> str:
162         if self.current + 1 >= len(self.source):
163             return "\0"
164         return self.source[self.current + 1]
165
166     def _is_at_end(self) -> bool:
167         return self.current >= len(self.source)
168
169     @staticmethod
170     def _is_identifier_start(char: str) -> bool:
171         return char.isalpha() or char == "_"
172
173     @staticmethod
174     def _is_identifier_part(char: str) -> bool:
175         return char.isalnum() or char == "_"
176
177     def _error_message(self, message: str) -> str:
178         return f"[line {self.line}, col {self.token_column}] {message}"
179
180     def iter_tokens(source: str) -> Iterator[Token]:
181         yield from Lexer(source).tokenize()

```

Listing 8.4: components/lexica.py — lexical analyser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from typing import Optional

```

```

5
6
7 #
8
9 #
10
11 @dataclass
12 class ASTNode:
13     """Base class for every AST node. Carries source location for
14         diagnostics."""
15     line: int = field(default=0, kw_only=True)
16     column: int = field(default=0, kw_only=True)
17
18 #
19
20 #
21
22 @dataclass
23 class Literal(ASTNode):
24     """An integer, float, boolean, or string literal.
25
26     Examples: 42    3.14    true    "hello"
27     """
28     value: int | float | bool | str
29
30
31 @dataclass
32 class Identifier(ASTNode):
33     """A variable or function name used as an expression.
34
35     Example: x
36     """
37     name: str
38
39
40 @dataclass
41 class BinaryOp(ASTNode):
42     """An arithmetic or comparison binary operation.
43
44     op is one of: '+' '-' '*' '/' '==' '!='
45     Example: a + b    x == 0
46     """
47     op: str
48     left: ASTNode

```

```

49     right: ASTNode
50
51
52 @dataclass
53 class FunctionCall(ASTNode):
54     """A call to a user-defined function.
55
56     Example:  add(1, 2)
57     """
58     name: str
59     args: list[ASTNode] = field(default_factory=list)
60
61
62 #
63 # -----
64 # Statements
65 # -----
66
67 @dataclass
68 class Assign(ASTNode):
69     """Variable assignment (first use infers type; later uses must
70     match).
71
72     Example:  x = 10;
73     """
74     name: str
75     value: ASTNode
76
77
78 @dataclass
79 class Print(ASTNode):
80     """Built-in print statement.
81
82     Example:  print(x);
83     """
84     expr: ASTNode
85
86
87 @dataclass
88 class Return(ASTNode):
89     """Return statement inside a function body.
90
91     Example:  return result;
92     """
93     expr: ASTNode
94
95
96 @dataclass
97 class Block(ASTNode):
98     """A brace-enclosed sequence of statements.

```

```

97
98     Example:  { x = 1; print(x); }
99     """
100     statements: list[ASTNode] = field(default_factory=list)
101
102
103 @dataclass
104 class If(ASTNode):
105     """If / optional else control flow.
106
107     Example:  if (x != 0) { print(x); } else { print(0); }
108     """
109     condition: ASTNode
110     then_block: Block
111     else_block: Optional[Block]
112
113
114 @dataclass
115 class While(ASTNode):
116     """While loop.
117
118     Example:  while (x != 0) { x = x - 1; }
119     """
120     condition: ASTNode
121     body: Block
122
123
124 @dataclass
125 class FunctionDef(ASTNode):
126     """Function definition.
127
128     Example:  def add(a, b) { return a + b; }
129
130     Note: parameter types are unknown at parse time -- Member 3's type
131           checker
132           will infer them from usage. We store only the parameter names here.
133     """
134     name: str
135     params: list[str]
136     body: Block
137
138 #
139 # -----
140 # Top-level
141 # -----
142
143 @dataclass
144 class Program(ASTNode):
145     """Root node: the entire source file as a list of statements."""

```

```
145 statements: list[ASTNode] = field(default_factory=list)
```

Listing 8.5: components/ast_nodes.py — AST node definitions

```
1 from __future__ import annotations
2
3 from typing import Optional
4
5 from components.tokens import Token, TokenType
6 from components.ast_nodes import (
7     ASTNode,
8     Program,
9     Block,
10    Assign,
11    If,
12    While,
13    FunctionDef,
14    FunctionCall,
15    Print,
16    Return,
17    BinaryOp,
18    Literal,
19    Identifier,
20 )
21
22
23 #
24 # -----
25
26 # Parser error
27 #
28 # -----
29
30
31 class ParseError(Exception):
32     pass
33
34 #
35 # -----
36
37 # Parser
38 #
39 # -----
40
41 class Parser:
42     """Recursive-descent parser.
43
44     Usage:
45         tokens = Lexer(source).tokenize()
46         tree = Parser(tokens).parse() # returns Program node
47     """
```

```

42
43 def __init__(self, tokens: list[Token]) -> None:
44     self._tokens = tokens
45     self._pos = 0
46
47     #
48     -----
49
50     # Public entry point
51     #
52     -----
53
54 def parse(self) -> Program:
55     """Parse the full token stream and return the root Program node
56     ."""
57     line, col = self._peek().line, self._peek().column
58     statements: list[ASTNode] = []
59     while not self._is_at_end():
60         statements.append(self._statement())
61     return Program(statements=statements, line=line, column=col)
62
63     #
64     -----
65
66     # Statements
67     #
68     -----
69
70
71 def _statement(self) -> ASTNode:
72     """Dispatch to the correct statement parser based on the
73     current token."""
74     tok = self._peek()
75
76     if tok.token_type == TokenType.DEF:
77         return self._func_def()
78
79     if tok.token_type == TokenType.IF:
80         return self._if_stmt()
81
82     if tok.token_type == TokenType.WHILE:
83         return self._while_stmt()
84
85     if tok.token_type == TokenType.RETURN:
86         return self._return_stmt()
87
88     if tok.token_type == TokenType.PRINT:
89         return self._print_stmt()
90
91     # assignment: IDENTIFIER "=" expr ";"
92     if (
93         tok.token_type == TokenType.IDENTIFIER

```

```

85         and self._peek_next().token_type == TokenType.ASSIGN
86     ):
87         return self._assignment()
88
89     # fallback: bare expression statement expr ";"
90     node = self._expr()
91     self._expect(TokenType.SEMICOLON, "Expected ';' after
92         expression")
93     return node
94
95 def _assignment(self) -> Assign:
96     """IDENTIFIER '=' expr ';'"""
97     name_tok = self._advance() #
98     IDENTIFIER
99     self._advance() # '='
100     value = self._expr()
101     self._expect(TokenType.SEMICOLON, "Expected ';' after
102         assignment")
103     return Assign(name=name_tok.lexeme, value=value,
104                   line=name_tok.line, column=name_tok.column)
105
106 def _if_stmt(self) -> If:
107     """'if' '(' bool_expr ')' block ( 'else' block )?"""
108     tok = self._advance() # 'if'
109     self._expect(TokenType.LPAREN, "Expected '(' after 'if'")
110     condition = self._bool_expr()
111     self._expect(TokenType.RPAREN, "Expected ')' after if condition
112         ")
113     then_block = self._block()
114     else_block: Optional[Block] = None
115     if self._check(TokenType.ELSE):
116         self._advance() # 'else'
117         else_block = self._block()
118     return If(condition=condition, then_block=then_block,
119              else_block=else_block,
120              line=tok.line, column=tok.column)
121
122 def _while_stmt(self) -> While:
123     """'while' '(' bool_expr ')' block"""
124     tok = self._advance() # '
125     while'
126     self._expect(TokenType.LPAREN, "Expected '(' after 'while'")
127     condition = self._bool_expr()
128     self._expect(TokenType.RPAREN, "Expected ')' after while
129         condition")
130     body = self._block()
131     return While(condition=condition, body=body,
132                  line=tok.line, column=tok.column)
133
134 def _func_def(self) -> FunctionDef:
135     """'def' IDENTIFIER '(' params? ')' block"""
136     tok = self._advance() # 'def'

```

```

130     name_tok = self._expect(TokenType.IDENTIFIER, "Expected
        function name after 'def'")
131     self._expect(TokenType.LPAREN, "Expected '(' after function
        name")
132     params = self._params()
133     self._expect(TokenType.RPAREN, "Expected ')' after parameters")
134     body = self._block()
135     return FunctionDef(name=name_tok.lexeme, params=params, body=
        body,
136                           line=tok.line, column=tok.column)
137
138 def _print_stmt(self) -> Print:
139     """'print' '(' expr ')' ';'"""
140     tok = self._advance() # '
        print'
141     self._expect(TokenType.LPAREN, "Expected '(' after 'print'")
142     expr = self._expr()
143     self._expect(TokenType.RPAREN, "Expected ')' after print
        expression")
144     self._expect(TokenType.SEMICOLON, "Expected ';' after print
        statement")
145     return Print(expr=expr, line=tok.line, column=tok.column)
146
147 def _return_stmt(self) -> Return:
148     """'return' expr ';'"""
149     tok = self._advance() # '
        return'
150     expr = self._expr()
151     self._expect(TokenType.SEMICOLON, "Expected ';' after return
        value")
152     return Return(expr=expr, line=tok.line, column=tok.column)
153
154 def _block(self) -> Block:
155     """'{', statement* '}'"""
156     tok = self._expect(TokenType.LBRACE, "Expected '{'")
157     statements: list[ASTNode] = []
158     while not self._check(TokenType.RBRACE) and not self._is_at_end
        ():
159         statements.append(self._statement())
160     self._expect(TokenType.RBRACE, "Expected '}' to close block")
161     return Block(statements=statements, line=tok.line, column=tok.
        column)
162
163 #
        -----
164 # Parameters (function definition)
165 #
        -----
166
167 def _params(self) -> list[str]:
168     """IDENTIFIER (',' IDENTIFIER)* -- returns list of param names.

```

```

169         """
170         params: list[str] = []
171         if self._check(TokenType.IDENTIFIER):
172             params.append(self._advance().lexeme)
173             while self._check(TokenType.COMMA):
174                 self._advance() # ',', '
175                 tok = self._expect(TokenType.IDENTIFIER, "Expected
176                     parameter name after ','")
177                 params.append(tok.lexeme)
178             return params
179
180 #
181 -----
182
183 # Boolean expressions (lowest precedence, used in if/while
184 # conditions)
185 #
186 -----
187
188 def _bool_expr(self) -> ASTNode:
189     """expr ('==' | '!=') expr | BOOL_LITERAL"""
190     # bare boolean literal: true / false
191     if self._check(TokenType.BOOL_LITERAL):
192         tok = self._advance()
193         value = tok.lexeme == "true"
194         return Literal(value=value, line=tok.line, column=tok.
195             column)
196
197     left = self._expr()
198
199     if self._check(TokenType.EQUAL_EQUAL):
200         op_tok = self._advance()
201         right = self._expr()
202         return BinaryOp(op=="==", left=left, right=right,
203             line=op_tok.line, column=op_tok.column)
204
205     if self._check(TokenType.BANG_EQUAL):
206         op_tok = self._advance()
207         right = self._expr()
208         return BinaryOp(op!="=", left=left, right=right,
209             line=op_tok.line, column=op_tok.column)
210
211     # no comparison operator found -- return the expr as-is
212     # (the type checker will validate it's Boolean)
213     return left
214
215 #
216 -----
217
218 # Arithmetic expressions (standard precedence)
219 #
220 -----

```

```

211
212 def _expr(self) -> ASTNode:
213     """term (('+' | '-') term)*"""
214     node = self._term()
215     while self._check(TokenType.PLUS) or self._check(TokenType.
216         MINUS):
217         op_tok = self._advance()
218         right = self._term()
219         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
220             line=op_tok.line, column=op_tok.column)
221     return node
222
223 def _term(self) -> ASTNode:
224     """factor (('*' | '/') factor)*"""
225     node = self._factor()
226     while self._check(TokenType.STAR) or self._check(TokenType.
227         SLASH):
228         op_tok = self._advance()
229         right = self._factor()
230         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
231             line=op_tok.line, column=op_tok.column)
232     return node
233
234 def _factor(self) -> ASTNode:
235     """
236     '-' factor (unary negation, desugared to 0 - factor)
237     | INT_LITERAL | FLOAT_LITERAL | STRING_LITERAL | BOOL_LITERAL
238     | IDENTIFIER ( '(' args? ') ' )?
239     | '(' expr ')'
240     """
241     tok = self._peek()
242
243     # Unary minus '-' factor
244     if tok.token_type == TokenType.MINUS:
245         op_tok = self._advance()
246         operand = self._factor()
247         zero = Literal(value=0, line=op_tok.line, column=op_tok.
248             column)
249         return BinaryOp(op="-", left=zero, right=operand,
250             line=op_tok.line, column=op_tok.column)
251
252     # Integer literal
253     if tok.token_type == TokenType.INT_LITERAL:
254         self._advance()
255         return Literal(value=int(tok.lexeme), line=tok.line, column
256             =tok.column)
257
258     # Float literal
259     if tok.token_type == TokenType.FLOAT_LITERAL:
260         self._advance()
261         return Literal(value=float(tok.lexeme), line=tok.line,
262             column=tok.column)

```

```

258
259 # Boolean literal
260 if tok.token_type == TokenType.BOOL_LITERAL:
261     self._advance()
262     return Literal(value=(tok.lexeme == "true"), line=tok.line,
263                     column=tok.column)
264
265 # String literal (keep the surrounding quotes stripped)
266 if tok.token_type == TokenType.STRING_LITERAL:
267     self._advance()
268     return Literal(value=tok.lexeme[1:-1], line=tok.line,
269                     column=tok.column)
270
271 # Identifier -- could be a plain variable or a function call
272 if tok.token_type == TokenType.IDENTIFIER:
273     self._advance()
274     if self._check(TokenType.LPAREN):
275         # function call
276         self._advance() # '('
277         args = self._args()
278         self._expect(TokenType.RPAREN, "Expected ')' after
279                     function arguments")
280         return FunctionCall(name=tok.lexeme, args=args,
281                             line=tok.line, column=tok.column)
282     return Identifier(name=tok.lexeme, line=tok.line, column=
283                       tok.column)
284
285 # Grouped expression '(' expr ')'
286 if tok.token_type == TokenType.LPAREN:
287     self._advance() # '('
288     node = self._expr()
289     self._expect(TokenType.RPAREN, "Expected ')' to close
290                     grouped expression")
291     return node
292
293     raise ParseError(self._error_at(tok, f"Unexpected token: {tok.
294                                     lexeme!r}"))
295
296 #
297 -----
298
299 # Arguments (function call)
300 #
301 -----
302
303 def _args(self) -> list[ASTNode]:
304     """expr (',' expr)*"""
305     args: list[ASTNode] = []
306     if not self._check(TokenType.RPAREN):
307         args.append(self._expr())
308         while self._check(TokenType.COMMA):
309             self._advance() # ','

```

```

301         args.append(self._expr())
302     return args
303
304     #
305     -----
306
307     # Token navigation helpers
308     #
309     -----
310
311     def _peek(self) -> Token:
312         return self._tokens[self._pos]
313
314     def _peek_next(self) -> Token:
315         if self._pos + 1 < len(self._tokens):
316             return self._tokens[self._pos + 1]
317         return self._tokens[-1] # EOF
318
319     def _advance(self) -> Token:
320         tok = self._tokens[self._pos]
321         if not self._is_at_end():
322             self._pos += 1
323         return tok
324
325     def _check(self, token_type: TokenType) -> bool:
326         return self._peek().token_type == token_type
327
328     def _is_at_end(self) -> bool:
329         return self._peek().token_type == TokenType.EOF
330
331     def _expect(self, token_type: TokenType, message: str) -> Token:
332         if self._check(token_type):
333             return self._advance()
334         raise ParseError(self._error_at(self._peek(), message))
335
336     def _error_at(self, tok: Token, message: str) -> str:
337         return f"[line {tok.line}, col {tok.column}] ParseError: {message}"

```

Listing 8.6: components/parser.py — recursive-descent parser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from enum import Enum
5
6
7 class LanguageType(Enum):
8     INTEGER = "Integer"
9     FLOAT = "Float"
10    BOOLEAN = "Boolean"
11    STRING = "String"

```

```

12     VOID = "Void"
13
14
15 @dataclass
16 class Symbol:
17     name: str
18     symbol_type: LanguageType
19
20
21 @dataclass
22 class VariableSymbol(Symbol):
23     initialized: bool = False
24
25
26 @dataclass
27 class FunctionSymbol(Symbol):
28     parameters: list[tuple[str, LanguageType]] = field(default_factory=
        list)
29
30
31 class SymbolTableError(Exception):
32     pass
33
34
35 class SymbolTable:
36     def __init__(self, parent: SymbolTable | None = None) -> None:
37         self.parent = parent
38         self._symbols: dict[str, Symbol] = {}
39
40     def define_variable(self, name: str, symbol_type: LanguageType,
41         initialized: bool = False) -> VariableSymbol:
42         if name in self._symbols:
43             raise SymbolTableError(f"Symbol already defined in this
44                 scope: {name}")
45         symbol = VariableSymbol(name=name, symbol_type=symbol_type,
46             initialized=initialized)
47         self._symbols[name] = symbol
48         return symbol
49
50     def define_function(
51         self,
52         name: str,
53         return_type: LanguageType,
54         parameters: list[tuple[str, LanguageType]],
55     ) -> FunctionSymbol:
56         if name in self._symbols:
57             raise SymbolTableError(f"Symbol already defined in this
58                 scope: {name}")
59         symbol = FunctionSymbol(name=name, symbol_type=return_type,
60             parameters=parameters)
61         self._symbols[name] = symbol
62         return symbol

```

```

59 def lookup(self, name: str) -> Symbol:
60     symbol = self._symbols.get(name)
61     if symbol is not None:
62         return symbol
63     if self.parent is not None:
64         return self.parent.lookup(name)
65     raise SymbolTableError(f"Undefined symbol: {name}")
66
67 def exists_in_current_scope(self, name: str) -> bool:
68     return name in self._symbols
69
70 def child_scope(self) -> SymbolTable:
71     return SymbolTable(parent=self)
72
73 def set_initialized(self, name: str) -> None:
74     symbol = self.lookup(name)
75     if not isinstance(symbol, VariableSymbol):
76         raise SymbolTableError(f"Cannot initialize non-variable
77                                 symbol: {name}")
78     symbol.initialized = True
79
80 def snapshot(self) -> dict[str, Symbol]:
81     return dict(self._symbols)
82
83 def format_table(self) -> str:
84     lines = [
85         f"{'Name':<12} {'Kind':<10} {'Type':<10} {'Initialized'
86             ':<12} Parameters",
87         "-" * 72,
88     ]
89
90     for name, symbol in self._symbols.items():
91         if isinstance(symbol, FunctionSymbol):
92             params = ", ".join(f"{param_name}: {param_type.value}"
93                                 for param_name, param_type in symbol.parameters)
94             lines.append(
95                 f"{name:<12} {'function':<10} {symbol.symbol_type.
96                     value:<10} {'-':<12} {params or '-'}"
97             )
98         elif isinstance(symbol, VariableSymbol):
99             initialized = "yes" if symbol.initialized else "no"
100             lines.append(
101                 f"{name:<12} {'variable':<10} {symbol.symbol_type.
102                     value:<10} {initialized:<12} -"
103             )
104         else:
105             lines.append(f"{name:<12} {'symbol':<10} {symbol.
106                             symbol_type.value:<10} {'-':<12} -")
107
108     return "\n".join(lines)

```

Listing 8.7: components/symbol_table.py — scoped symbol table

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4
5 from components.ast_nodes import (
6     ASTNode,
7     Assign,
8     BinaryOp,
9     Block,
10    FunctionCall,
11    FunctionDef,
12    Identifier,
13    If,
14    Literal,
15    Print,
16    Program,
17    Return,
18    While,
19 )
20 from components.symbol_table import FunctionSymbol, LanguageType,
21     SymbolTable, SymbolTableError, VariableSymbol
22
23 class TypeCheckError(Exception):
24     pass
25
26
27 @dataclass
28 class _FunctionState:
29     node: FunctionDef
30     body_checked: bool = False
31
32
33 @dataclass
34 class _FunctionContext:
35     name: str
36     return_type: LanguageType | None = None
37
38
39 class TypeChecker:
40     def __init__(self) -> None:
41         self.global_scope = SymbolTable()
42         self._functions: dict[str, _FunctionState] = {}
43         self._active_calls: list[str] = []
44
45     def check(self, program: Program) -> SymbolTable:
46         for statement in program.statements:
47             if isinstance(statement, FunctionDef):
48                 self._register_function(statement)
49
50         for statement in program.statements:
51             if isinstance(statement, FunctionDef):

```

```

52         continue
53         self._check_statement(statement, self.global_scope, None)
54
55     self._check_uncalled_functions()
56     return self.global_scope
57
58     def _register_function(self, node: FunctionDef) -> None:
59         if node.name in self._functions:
60             raise TypeCheckError(self._error_at(node, f"Function '{node
61                 .name}' is already defined"))
62
63         placeholder_parameters = [(param_name, LanguageType.VOID) for
64             param_name in node.params]
65         try:
66             self.global_scope.define_function(node.name, LanguageType.
67                 VOID, placeholder_parameters)
68         except SymbolTableError as error:
69             raise TypeCheckError(self._error_at(node, str(error))) from
70                 error
71         self._functions[node.name] = _FunctionState(node=node)
72
73     def _check_statement(
74         self,
75         node: ASTNode,
76         scope: SymbolTable,
77         function_context: _FunctionContext | None,
78     ) -> None:
79         if isinstance(node, Assign):
80             value_type = self._infer_expr_type(node.value, scope)
81             self._assign_variable_type(scope, node, value_type)
82             return
83
84         if isinstance(node, Print):
85             self._infer_expr_type(node.expr, scope)
86             return
87
88         if isinstance(node, Return):
89             if function_context is None:
90                 raise TypeCheckError(self._error_at(node, "'return' is
91                     only valid inside a function"))
92             expr_type = self._infer_expr_type(node.expr, scope)
93             if function_context.return_type is None:
94                 function_context.return_type = expr_type
95             elif function_context.return_type != expr_type:
96                 raise TypeCheckError(
97                     self._error_at(
98                         node,
99                         f"Function '{function_context.name}' returns
100                             both {function_context.return_type.value}
101                             and {expr_type.value}",
102                     )
103             )
104         )
105     )
106     return

```

```

98
99     if isinstance(node, If):
100         condition_type = self._infer_expr_type(node.condition,
101                                                 scope)
102         if condition_type != LanguageType.BOOLEAN:
103             raise TypeCheckError(self._error_at(node.condition, "If
104                                     condition must have type Boolean"))
105         self._check_block(node.then_block, scope.child_scope(),
106                           function_context)
107         if node.else_block is not None:
108             self._check_block(node.else_block, scope.child_scope(),
109                               function_context)
110         return
111
112     if isinstance(node, While):
113         condition_type = self._infer_expr_type(node.condition,
114                                                 scope)
115         if condition_type != LanguageType.BOOLEAN:
116             raise TypeCheckError(self._error_at(node.condition, "
117                                     While condition must have type Boolean"))
118         self._check_block(node.body, scope.child_scope(),
119                           function_context)
120         return
121
122     if isinstance(node, Block):
123         self._check_block(node, scope.child_scope(),
124                           function_context)
125         return
126
127     if isinstance(node, FunctionCall):
128         self._infer_function_call_type(node, scope)
129         return
130
131     raise TypeCheckError(self._error_at(node, f"Unsupported
132                                     statement node: {type(node).__name__}"))
133
134 def _check_block(
135     self,
136     block: Block,
137     scope: SymbolTable,
138     function_context: _FunctionContext | None,
139 ) -> None:
140     for statement in block.statements:
141         self._check_statement(statement, scope, function_context)
142
143 def _infer_expr_type(self, node: ASTNode, scope: SymbolTable) ->
144 LanguageType:
145     if isinstance(node, Literal):
146         return self._literal_type(node)
147
148     if isinstance(node, Identifier):
149         try:
150             symbol = scope.lookup(node.name)

```

```

141         except SymbolTableError as error:
142             raise TypeCheckError(self._error_at(node, str(error)))
143             from error
144         if not isinstance(symbol, VariableSymbol):
145             raise TypeCheckError(self._error_at(node, f"'{node.name
146             }' is not a variable"))
147         return symbol.symbol_type
148
149     if isinstance(node, FunctionCall):
150         return self._infer_function_call_type(node, scope)
151
152     if isinstance(node, BinaryOp):
153         left_type = self._infer_expr_type(node.left, scope)
154         right_type = self._infer_expr_type(node.right, scope)
155         return self._infer_binary_type(node, left_type, right_type)
156
157     raise TypeCheckError(self._error_at(node, f"Unsupported
158     expression node: {type(node).__name__}"))
159
160 def _infer_function_call_type(self, node: FunctionCall, scope:
161 SymbolTable) -> LanguageType:
162     function_state = self._functions.get(node.name)
163     if function_state is None:
164         raise TypeCheckError(self._error_at(node, f"Undefined
165         function: {node.name}"))
166     if node.name in self._active_calls:
167         raise TypeCheckError(self._error_at(node, f"Recursive
168         function calls are not supported: {node.name}"))
169
170     argument_types = [self._infer_expr_type(argument, scope) for
171     argument in node.args]
172     function_symbol = self._lookup_function_symbol(node)
173
174     if len(argument_types) != len(function_symbol.parameters):
175         raise TypeCheckError(
176             self._error_at(
177                 node,
178                 f"Function '{node.name}' expects {len(
179                     function_symbol.parameters)} argument(s), got {
180                     len(argument_types)}",
181             )
182         )
183
184     declared_parameter_types = [param_type for _, param_type in
185     function_symbol.parameters]
186     if all(param_type == LanguageType.VOID for param_type in
187     declared_parameter_types):
188         function_symbol.parameters = list(zip(function_state.node.
189         params, argument_types, strict=False))
190     else:
191         for index, ((_, expected_type), actual_type) in enumerate(
192             zip(function_symbol.parameters, argument_types, strict=
193             False)):

```

```

180         if expected_type != actual_type:
181             raise TypeCheckError(
182                 self._error_at(
183                     node.args[index],
184                     f"Argument {index + 1} of '{node.name}'
                        must be {expected_type.value}, got {
                        actual_type.value}",
185                 )
186             )
187
188     if not function_state.body_checked:
189         self._check_function_body(function_state, function_symbol)
190
191     return function_symbol.symbol_type
192
193 def _check_function_body(self, function_state: _FunctionState,
194 function_symbol: FunctionSymbol) -> None:
195     function_context = _FunctionContext(name=function_state.node.
196         name)
197     function_scope = self.global_scope.child_scope()
198     for parameter_name, parameter_type in function_symbol.
199         parameters:
200         function_scope.define_variable(parameter_name,
201             parameter_type, initialized=True)
202
203     self._active_calls.append(function_state.node.name)
204     try:
205         self._check_block(function_state.node.body, function_scope,
206             function_context)
207     finally:
208         self._active_calls.pop()
209
210     function_symbol.symbol_type = function_context.return_type or
211         LanguageType.VOID
212     function_state.body_checked = True
213
214 def _check_uncalled_functions(self) -> None:
215     """After all call-site checks, type-check any function whose
216         body was never visited."""
217     for name, state in self._functions.items():
218         if not state.body_checked:
219             try:
220                 function_symbol = self.global_scope.lookup(name)
221             except SymbolTableError:
222                 continue
223             if not isinstance(function_symbol, FunctionSymbol):
224                 continue
225             inferred_params = self._infer_param_types_from_body(
226                 state.node)
227             function_symbol.parameters = inferred_params
228             self._check_function_body(state, function_symbol)
229
230 def _infer_param_types_from_body(self, node: FunctionDef) -> list[

```

```

223 tuple[str, LanguageType]]:
224     """Infer parameter types by scanning the body for expression
225     contexts.
226
227     Any parameter used as an operand of an arithmetic or comparison
228     operator
229     is inferred as numeric (Float if the sibling operand is a float
230     literal,
231     Integer otherwise). Parameters whose usage gives no type hint
232     default to
233     Integer.
234     """
235     param_names = set(node.params)
236     inferred: dict[str, LanguageType] = {}
237
238     def scan_expr(n: ASTNode) -> None:
239         if isinstance(n, BinaryOp):
240             if n.op in {"+", "-", "*", "/", "=", "!="}:
241                 for operand, other in ((n.left, n.right), (n.right,
242                     n.left)):
243                     if isinstance(operand, Identifier) and operand.
244                         name in param_names:
245                         if isinstance(other, Literal) and
246                             isinstance(other.value, float):
247                             inferred.setdefault(operand.name,
248                                 LanguageType.FLOAT)
249                         else:
250                             inferred.setdefault(operand.name,
251                                 LanguageType.INTEGER)
252                     scan_expr(n.left)
253                     scan_expr(n.right)
254             elif isinstance(n, FunctionCall):
255                 for arg in n.args:
256                     scan_expr(arg)
257
258     def scan_stmt(n: ASTNode) -> None:
259         if isinstance(n, Assign):
260             scan_expr(n.value)
261         elif isinstance(n, Print):
262             scan_expr(n.expr)
263         elif isinstance(n, Return):
264             scan_expr(n.expr)
265         elif isinstance(n, If):
266             scan_expr(n.condition)
267             scan_block(n.then_block)
268             if n.else_block:
269                 scan_block(n.else_block)
270         elif isinstance(n, While):
271             scan_expr(n.condition)
272             scan_block(n.body)
273         elif isinstance(n, Block):
274             scan_block(n)
275         elif isinstance(n, FunctionCall):

```

```

266         for arg in n.args:
267             scan_expr(arg)
268
269     def scan_block(block: Block) -> None:
270         for stmt in block.statements:
271             scan_stmt(stmt)
272
273     scan_block(node.body)
274     return [(param, inferred.get(param, LanguageType.INTEGER)) for
275             param in node.params]
276
277 def _assign_variable_type(self, scope: SymbolTable, node: Assign,
278 value_type: LanguageType) -> None:
279     try:
280         symbol = scope.lookup(node.name)
281     except SymbolTableError:
282         scope.define_variable(node.name, value_type, initialized=
283                               True)
284     return
285
286     if not isinstance(symbol, VariableSymbol):
287         raise TypeCheckError(self._error_at(node, f"'{node.name}'
288         is not a variable"))
289     if symbol.symbol_type != value_type:
290         raise TypeCheckError(
291             self._error_at(
292                 node,
293                 f"Cannot assign {value_type.value} to variable '{
294                 node.name}' of type {symbol.symbol_type.value}",
295             )
296         )
297     symbol.initialized = True
298
299 def _infer_binary_type(
300     self,
301     node: BinaryOp,
302     left_type: LanguageType,
303     right_type: LanguageType,
304 ) -> LanguageType:
305     if node.op in {"+", "-", "*", "/"}:
306         if not self._is_numeric(left_type) or not self._is_numeric(
307             right_type):
308             raise TypeCheckError(self._error_at(node, f"Operator '{
309             node.op}' requires numeric operands"))
310         if node.op == "/":
311             return LanguageType.FLOAT
312         if left_type == LanguageType.FLOAT or right_type ==
313             LanguageType.FLOAT:
314             return LanguageType.FLOAT
315         return LanguageType.INTEGER
316
317     if node.op in {"==", "!="}:
318         numeric_comparison = self._is_numeric(left_type) and self.

```

```

        _is_numeric(right_type)
311     if not numeric_comparison:
312         raise TypeCheckError(
313             self._error_at(
314                 node,
315                 f"Operator '{node.op}' requires two arithmetic
                    operands",
316             )
317         )
318     return LanguageType.BOOLEAN
319
320     raise TypeCheckError(self._error_at(node, f"Unsupported
        operator: {node.op}"))
321
322 def _lookup_function_symbol(self, node: FunctionCall) ->
    FunctionSymbol:
323     try:
324         symbol = self.global_scope.lookup(node.name)
325     except SymbolTableError as error:
326         raise TypeCheckError(self._error_at(node, str(error))) from
            error
327     if not isinstance(symbol, FunctionSymbol):
328         raise TypeCheckError(self._error_at(node, f"'{node.name}'
            is not a function"))
329     return symbol
330
331 @staticmethod
332 def _is_numeric(language_type: LanguageType) -> bool:
333     return language_type in {LanguageType.INTEGER, LanguageType.
        FLOAT}
334
335 @staticmethod
336 def _literal_type(node: Literal) -> LanguageType:
337     if isinstance(node.value, bool):
338         return LanguageType.BOOLEAN
339     if isinstance(node.value, int):
340         return LanguageType.INTEGER
341     if isinstance(node.value, float):
342         return LanguageType.FLOAT
343     if isinstance(node.value, str):
344         return LanguageType.STRING
345     raise TypeCheckError(f"Unsupported literal value: {node.value!r}
        ")
346
347 @staticmethod
348 def _error_at(node: ASTNode, message: str) -> str:
349     return f"[line {node.line}, col {node.column}] TypeError: {
        message}"

```

Listing 8.8: components/type_checker.py — static type checker with inference

```

1 from __future__ import annotations
2

```

```

3 from dataclasses import dataclass
4 from typing import Callable
5
6 from components.ast_nodes import (
7     ASTNode,
8     Assign,
9     BinaryOp,
10    Block,
11    FunctionCall,
12    FunctionDef,
13    Identifier,
14    If,
15    Literal,
16    Print,
17    Program,
18    Return,
19    While,
20 )
21 from components.symbol_table import LanguageType
22
23
24 class InterpreterError(Exception):
25     pass
26
27
28 class _ReturnSignal(Exception):
29     def __init__(self, value: object) -> None:
30         self.value = value
31
32
33 @dataclass
34 class _FunctionRuntime:
35     node: FunctionDef
36
37
38 class _Environment:
39     def __init__(self, parent: _Environment | None = None) -> None:
40         self.parent = parent
41         self.values: dict[str, object] = {}
42
43     def define(self, name: str, value: object) -> None:
44         self.values[name] = value
45
46     def get(self, name: str) -> object:
47         if name in self.values:
48             return self.values[name]
49         if self.parent is not None:
50             return self.parent.get(name)
51         raise InterpreterError(f"Undefined variable: {name}")
52
53     def assign(self, name: str, value: object) -> None:
54         if name in self.values:
55             self.values[name] = value

```

```

56         return
57     if self.parent is not None and self.parent.contains(name):
58         self.parent.assign(name, value)
59         return
60     self.values[name] = value
61
62     def contains(self, name: str) -> bool:
63         if name in self.values:
64             return True
65         if self.parent is not None:
66             return self.parent.contains(name)
67         return False
68
69
70 class Interpreter:
71     def __init__(self, output_callback: Callable[[str], None] | None =
72         None) -> None:
73         self._functions: dict[str, _FunctionRuntime] = {}
74         self._output_callback = output_callback or print
75
76     def execute(self, program: Program) -> None:
77         environment = _Environment()
78         for statement in program.statements:
79             if isinstance(statement, FunctionDef):
80                 self._functions[statement.name] = _FunctionRuntime(node
81                     =statement)
82
83         for statement in program.statements:
84             if isinstance(statement, FunctionDef):
85                 continue
86             self._execute_statement(statement, environment)
87
88     def _execute_statement(self, node: ASTNode, environment:
89         _Environment) -> None:
90         if isinstance(node, Assign):
91             value = self._evaluate(node.value, environment)
92             environment.assign(node.name, value)
93             return
94
95         if isinstance(node, Print):
96             value = self._evaluate(node.expr, environment)
97             self._output_callback(f"{self._format_value(value)} : {self
98                 ._runtime_type(value).value}")
99             return
100
101         if isinstance(node, Return):
102             raise _ReturnSignal(self._evaluate(node.expr, environment))
103
104         if isinstance(node, If):
105             condition = self._evaluate(node.condition, environment)
106             if not isinstance(condition, bool):
107                 raise InterpreterError(self._error_at(node.condition, "
108                     If condition must evaluate to Boolean"))

```

```

104         branch = node.then_block if condition else node.else_block
105         if branch is not None:
106             self._execute_block(branch, _Environment(parent=
107                 environment))
108         return
109
110     if isinstance(node, While):
111         while True:
112             condition = self._evaluate(node.condition, environment)
113             if not isinstance(condition, bool):
114                 raise InterpreterError(self._error_at(node,
115                     condition, "While condition must evaluate to
116                         Boolean"))
117             if not condition:
118                 break
119             self._execute_block(node.body, _Environment(parent=
120                 environment))
121         return
122
123     if isinstance(node, Block):
124         self._execute_block(node, _Environment(parent=environment))
125         return
126
127     if isinstance(node, FunctionCall):
128         self._evaluate(node, environment)
129         return
130
131     raise InterpreterError(self._error_at(node, f"Unsupported
132         statement node: {type(node).__name__}"))
133
134 def _execute_block(self, block: Block, environment: _Environment)
135     -> None:
136     for statement in block.statements:
137         self._execute_statement(statement, environment)
138
139 def _evaluate(self, node: ASTNode, environment: _Environment) ->
140     object:
141     if isinstance(node, Literal):
142         return node.value
143
144     if isinstance(node, Identifier):
145         return environment.get(node.name)
146
147     if isinstance(node, BinaryOp):
148         left_value = self._evaluate(node.left, environment)
149         right_value = self._evaluate(node.right, environment)
150         return self._evaluate_binary(node, left_value, right_value)
151
152     if isinstance(node, FunctionCall):
153         return self._call_function(node, environment)
154
155     raise InterpreterError(self._error_at(node, f"Unsupported
156         expression node: {type(node).__name__}"))

```

```

149
150 def _call_function(self, node: FunctionCall, environment:
    _Environment) -> object:
151     function_runtime = self._functions.get(node.name)
152     if function_runtime is None:
153         raise InterpreterError(self._error_at(node, f"Undefined
            function: {node.name}"))
154
155     if len(node.args) != len(function_runtime.node.params):
156         raise InterpreterError(
157             self._error_at(
158                 node,
159                 f"Function '{node.name}' expects {len(
                    function_runtime.node.params)} argument(s), got
                    {len(node.args)}",
160             )
161         )
162
163     call_environment = _Environment(parent=environment)
164     argument_values = [self._evaluate(argument, environment) for
        argument in node.args]
165     for parameter_name, argument_value in zip(function_runtime.node
        .params, argument_values, strict=False):
166         call_environment.define(parameter_name, argument_value)
167
168     try:
169         self._execute_block(function_runtime.node.body,
            call_environment)
170     except _ReturnSignal as signal:
171         return signal.value
172     return None
173
174 def _evaluate_binary(self, node: BinaryOp, left_value: object,
    right_value: object) -> object:
175     if node.op == "+":
176         return left_value + right_value
177     if node.op == "-":
178         return left_value - right_value
179     if node.op == "*":
180         return left_value * right_value
181     if node.op == "/":
182         return left_value / right_value
183     if node.op == "==":
184         return left_value == right_value
185     if node.op == "!=":
186         return left_value != right_value
187     raise InterpreterError(self._error_at(node, f"Unsupported
        operator: {node.op}"))
188
189 @staticmethod
190 def _format_value(value: object) -> str:
191     if isinstance(value, str):
192         return f'"{value}"'

```

```

193         if isinstance(value, bool):
194             return "true" if value else "false"
195         return str(value)
196
197     @staticmethod
198     def _runtime_type(value: object) -> LanguageType:
199         if isinstance(value, bool):
200             return LanguageType.BOOLEAN
201         if isinstance(value, int):
202             return LanguageType.INTEGER
203         if isinstance(value, float):
204             return LanguageType.FLOAT
205         if isinstance(value, str):
206             return LanguageType.STRING
207         return LanguageType.VOID
208
209     @staticmethod
210     def _error_at(node: ASTNode, message: str) -> str:
211         return f"[line {node.line}, col {node.column}] RuntimeError: {
            message}"

```

Listing 8.9: components/interpreter.py — tree-walking interpreter

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass
4
5  from components.ast_printer import ASTPrinter
6  from components.interpreter import Interpreter
7  from components.lexica import Lexer
8  from components.parser import Parser
9  from components.symbol_table import SymbolTable
10 from components.tokens import Token
11 from components.type_checker import TypeChecker
12
13
14 @dataclass
15 class PipelineResult:
16     tokens: list[Token]
17     ast_text: str
18     checked_scope: SymbolTable
19     outputs: list[str]
20
21
22 def run_pipeline(source_code: str) -> PipelineResult:
23     tokens = Lexer(source_code).tokenize()
24     tree = Parser(tokens).parse()
25     ast_text = ASTPrinter().print(tree)
26     checked_scope = TypeChecker().check(tree)
27
28     outputs: list[str] = []
29     Interpreter(output_callback=outputs.append).execute(tree)
30

```

```

31     return PipelineResult(
32         tokens=tokens,
33         ast_text=ast_text,
34         checked_scope=checked_scope,
35         outputs=outputs,
36     )
37
38
39 def format_tokens(tokens: list[Token]) -> str:
40     lines = []
41     for token in tokens:
42         lines.append(
43             f"    {token.token_type.name:<14} {token.lexeme!r:<12} line={
44                 token.line} col={token.column}"
45         )
46     return "\n".join(lines)
47
48 def format_stage_output(result: PipelineResult) -> str:
49     sections = [
50         "=" * 60,
51         "    STAGE 1 -- TOKENS",
52         "=" * 60,
53         format_tokens(result.tokens),
54         "",
55         "=" * 60,
56         "    STAGE 2 -- AST (parser output)",
57         "=" * 60,
58         result.ast_text,
59         "",
60         "=" * 60,
61         "    STAGE 3 -- TYPE CHECK",
62         "=" * 60,
63         result.checked_scope.format_table(),
64         "",
65         "=" * 60,
66         "    STAGE 4 -- EXECUTION",
67         "=" * 60,
68         "\n".join(result.outputs) if result.outputs else "(no output)",
69     ]
70     return "\n".join(sections)

```

Listing 8.10: components/pipeline.py — shared pipeline (CLI, UI, and tests)

A.3 Test Suite

```

1 from __future__ import annotations
2
3 import unittest
4
5 from components.lexica import Lexer, LexerError
6 from components.parser import Parser, ParseError

```

```

7 from components.pipeline import run_pipeline
8 from components.symbol_table import LanguageType, SymbolTable,
  SymbolTableError
9 from components.tokens import TokenType
10 from components.type_checker import TypeCheckError
11
12
13 #
14 # -----
15 # Lexer
16 # -----
17
18 class LexerTests(unittest.TestCase):
19     """Tests for components/lexica.py -- lexical analysis and
20     tokenisation."""
21
22     def test_integer_literal_token(self) -> None:
23         tokens = Lexer("42").tokenize()
24         self.assertEqual(tokens[0].token_type, TokenType.INT_LITERAL)
25         self.assertEqual(tokens[0].lexeme, "42")
26
27     def test_float_literal_token(self) -> None:
28         tokens = Lexer("3.14").tokenize()
29         self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
30         self.assertEqual(tokens[0].lexeme, "3.14")
31
32     def test_bool_literals_produce_bool_literal_tokens(self) -> None:
33         tokens = Lexer("true false").tokenize()
34         self.assertEqual(tokens[0].token_type, TokenType.BOOL_LITERAL)
35         self.assertEqual(tokens[0].lexeme, "true")
36         self.assertEqual(tokens[1].token_type, TokenType.BOOL_LITERAL)
37         self.assertEqual(tokens[1].lexeme, "false")
38
39     def test_keywords_produce_correct_token_types(self) -> None:
40         tokens = Lexer("if else while def return print").tokenize()
41         expected = [
42             TokenType.IF, TokenType.ELSE, TokenType.WHILE,
43             TokenType.DEF, TokenType.RETURN, TokenType.PRINT,
44         ]
45         actual = [t.token_type for t in tokens if t.token_type !=
46                   TokenType.EOF]
47         self.assertEqual(actual, expected)
48
49     def test_identifier_not_confused_with_keyword(self) -> None:
50         tokens = Lexer("iffy whileloop").tokenize()
51         self.assertEqual(tokens[0].token_type, TokenType.IDENTIFIER)
52         self.assertEqual(tokens[1].token_type, TokenType.IDENTIFIER)
53
54     def test_two_character_operators(self) -> None:
55         tokens = Lexer("== !=").tokenize()

```

```

53     self.assertEqual(tokens[0].token_type, TokenType.EQUAL_EQUAL)
54     self.assertEqual(tokens[1].token_type, TokenType.BANG_EQUAL)
55
56     def test_source_position_tracked_across_lines(self) -> None:
57         tokens = Lexer("x\ny").tokenize()
58         self.assertEqual(tokens[0].line, 1)
59         self.assertEqual(tokens[1].line, 2)
60
61     def test_unexpected_character_raises_lexer_error(self) -> None:
62         with self.assertRaises(LexerError):
63             Lexer("@").tokenize()
64
65     def test_lone_exclamation_raises_lexer_error(self) -> None:
66         with self.assertRaises(LexerError):
67             Lexer("!5").tokenize()
68
69     def test_unterminated_string_raises_lexer_error(self) -> None:
70         with self.assertRaises(LexerError):
71             Lexer('"hello').tokenize()
72
73
74 #
75 # Symbol table
76 #
77
78 class SymbolTableTests(unittest.TestCase):
79     """Tests for components/symbol_table.py -- scoped symbol table and
80     semantic types."""
81
82     def test_define_variable_and_lookup(self) -> None:
83         table = SymbolTable()
84         table.define_variable("x", LanguageType.INTEGER, initialized=
85             True)
86         symbol = table.lookup("x")
87         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
88
89     def test_duplicate_variable_definition_raises_error(self) -> None:
90         table = SymbolTable()
91         table.define_variable("x", LanguageType.INTEGER)
92         with self.assertRaises(SymbolTableError):
93             table.define_variable("x", LanguageType.FLOAT)
94
95     def test_duplicate_function_definition_raises_error(self) -> None:
96         table = SymbolTable()
97         table.define_function("f", LanguageType.INTEGER, [])
98         with self.assertRaises(SymbolTableError):
99             table.define_function("f", LanguageType.FLOAT, [])
100
101     def test_lookup_in_parent_scope(self) -> None:

```

```

100     parent = SymbolTable()
101     parent.define_variable("x", LanguageType.INTEGER, initialized=
        True)
102     child = parent.child_scope()
103     symbol = child.lookup("x")
104     self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
105
106     def test_undefined_symbol_raises_error(self) -> None:
107         table = SymbolTable()
108         with self.assertRaises(SymbolTableError):
109             table.lookup("z")
110
111     def test_format_table_contains_variable_row(self) -> None:
112         table = SymbolTable()
113         table.define_variable("counter", LanguageType.INTEGER,
            initialized=True)
114         output = table.format_table()
115         self.assertIn("counter", output)
116         self.assertIn("variable", output)
117         self.assertIn("Integer", output)
118
119     def test_format_table_contains_function_row(self) -> None:
120         table = SymbolTable()
121         table.define_function("add", LanguageType.INTEGER, [("a",
            LanguageType.INTEGER)])
122         output = table.format_table()
123         self.assertIn("add", output)
124         self.assertIn("function", output)
125
126
127 #
128 # Parser
129 #
130
131 class ParserTests(unittest.TestCase):
132     """Tests for components/parser.py -- recursive-descent parsing."""
133
134     def _parse(self, source: str):
135         return Parser(Lexer(source).tokenize()).parse()
136
137     def test_missing_semicolon_raises_parse_error(self) -> None:
138         with self.assertRaises(ParseError):
139             self._parse("x = 5")
140
141     def test_unclosed_brace_raises_parse_error(self) -> None:
142         with self.assertRaises(ParseError):
143             self._parse("if (true) { x = 1;}")
144
145     def test_operator_precedence_mul_before_add(self) -> None:

```

```

146     # 2 + 3 * 4 must be 14, not 20
147     result = run_pipeline("x = 2 + 3 * 4; print(x);")
148     self.assertEqual(result.outputs, ["14 : Integer"])
149
150     def test_operator_left_associativity(self) -> None:
151         # 10 - 3 - 2 must be 5 (left-associ), not 9
152         result = run_pipeline("x = 10 - 3 - 2; print(x);")
153         self.assertEqual(result.outputs, ["5 : Integer"])
154
155     def test_parentheses_override_precedence(self) -> None:
156         # (2 + 3) * 4 must be 20
157         result = run_pipeline("x = (2 + 3) * 4; print(x);")
158         self.assertEqual(result.outputs, ["20 : Integer"])
159
160
161     #
162     -----
163
164     # Type checker
165     #
166     -----
167
168
169     class TypeCheckerTests(unittest.TestCase):
170         """Tests for components/type_checker.py -- static type inference and
171         checking."""
172
173         def test_integer_arithmetic_stays_integer(self) -> None:
174             result = run_pipeline("x = 3 + 4; print(x);")
175             self.assertEqual(result.outputs, ["7 : Integer"])
176
177         def test_division_always_produces_float(self) -> None:
178             # integer / integer -> Float by design
179             result = run_pipeline("x = 10 / 2; print(x);")
180             self.assertEqual(result.outputs, ["5.0 : Float"])
181
182         def test_float_promotion_in_addition(self) -> None:
183             result = run_pipeline("x = 1 + 2.5; print(x);")
184             self.assertEqual(result.outputs, ["3.5 : Float"])
185
186         def test_string_variable_type_inferred(self) -> None:
187             result = run_pipeline('x = "hello"; print(x);')
188             self.assertEqual(result.outputs, ['"hello" : String'])
189
190         def test_boolean_literal_type_inferred(self) -> None:
191             result = run_pipeline("x = true; print(x);")
192             self.assertEqual(result.outputs, ["true : Boolean"])
193
194         def test_boolean_equality_raises_type_error(self) -> None:
195             # Boolean == Boolean is not allowed; == only accepts arithmetic
196             operands
197             with self.assertRaises(TypeError):

```

```

192         run_pipeline('x = true; if (x == false) { print("then"); }
193             else { print("else"); }')
194
195 def test_arithmetic_on_string_raises_type_error(self) -> None:
196     with self.assertRaises(TypeError):
197         run_pipeline('x = "a" + 1;')
198
199 def test_if_non_boolean_condition_raises_type_error(self) -> None:
200     with self.assertRaises(TypeError):
201         run_pipeline("if (1) { x = 2; }")
202
203 def test_while_non_boolean_condition_raises_type_error(self) ->
204     None:
205     with self.assertRaises(TypeError):
206         run_pipeline("x = 0; while (x) { x = x + 1; }")
207
208 def test_assignment_type_mismatch_raises_type_error(self) -> None:
209     with self.assertRaises(TypeError):
210         run_pipeline('x = 5; x = "hello";')
211
212 def test_undefined_variable_raises_type_error(self) -> None:
213     with self.assertRaises(TypeError):
214         run_pipeline("print(z);")
215
216 def test_function_wrong_arg_count_raises_type_error(self) -> None:
217     with self.assertRaises(TypeError):
218         run_pipeline("def f(a) { return a; } f(1, 2);")
219
220 def test_function_arg_type_mismatch_raises_type_error(self) -> None
221 :
222     # first call infers a: Integer; second call with String must
223     fail
224     with self.assertRaises(TypeError):
225         run_pipeline('def f(a) { return a; } f(1); f("hello");')
226
227 def test_uncalled_function_body_type_error_is_caught(self) -> None:
228     # type error inside body must be detected even if function is
229     never called
230     with self.assertRaises(TypeError):
231         run_pipeline('def bad() { y = "hello" + 1; }')
232
233 def test_valid_uncalled_function_does_not_raise(self) -> None:
234     # a well-typed but uncalled function should not raise
235     result = run_pipeline("def add(a, b) { return a + b; }")
236     self.assertEqual(result.outputs, [])
237
238 #
239
240 # Integration (full pipeline)
241 #

```

```

237
238 class IntegrationTests(unittest.TestCase):
239     """End-to-end tests through all four pipeline stages."""
240
241     # --- Unary minus ---
242
243     def test_unary_minus_integer(self) -> None:
244         result = run_pipeline("x = -5; print(x);")
245         self.assertEqual(result.outputs, ["-5 : Integer"])
246
247     def test_unary_minus_float(self) -> None:
248         result = run_pipeline("y = -3.14; print(y);")
249         self.assertEqual(result.outputs, ["-3.14 : Float"])
250
251     def test_unary_minus_variable(self) -> None:
252         result = run_pipeline("x = 10; y = -x; print(y);")
253         self.assertEqual(result.outputs, ["-10 : Integer"])
254
255     def test_unary_minus_inside_expression(self) -> None:
256         result = run_pipeline("x = 3 + -2; print(x);")
257         self.assertEqual(result.outputs, ["1 : Integer"])
258
259     # --- Control flow ---
260
261     def test_if_then_branch_executes(self) -> None:
262         result = run_pipeline(
263             'x = 5; if (x == 5) { print("yes"); } else { print("no"); }
264             '
265         )
266         self.assertEqual(result.outputs, ['"yes" : String'])
267
268     def test_if_else_branch_executes_when_condition_false(self) -> None:
269         result = run_pipeline(
270             'x = 0; if (x == 5) { print("yes"); } else { print("no"); }
271             '
272         )
273         self.assertEqual(result.outputs, ['"no" : String'])
274
275     def test_while_loop_executes_repeatedly(self) -> None:
276         result = run_pipeline("x = 3; while (x != 0) { print(x); x = x
277             - 1; }")
278         self.assertEqual(result.outputs, ["3 : Integer", "2 : Integer",
279             "1 : Integer"])
280
281     # --- Functions ---
282
283     def test_function_integer_return(self) -> None:
284         result = run_pipeline("def square(n) { return n * n; } print(
285             square(7));")
286         self.assertEqual(result.outputs, ["49 : Integer"])

```

```

283 def test_function_type_inference(self) -> None:
284     result = run_pipeline(
285         "def add(a, b) { return a + b; } result = add(2, 3.5);
          print(result);"
286     )
287     self.assertEqual(result.outputs, ["5.5 : Float"])
288     self.assertIn("result          variable   Float", result.
          checked_scope.format_table())
289
290 def test_function_value_parameter_passing(self) -> None:
291     # Modifying a parameter inside a function must not affect the
292     caller's variable
293     result = run_pipeline(
294         "def inc(a) { a = a + 1; return a; } x = 10; y = inc(x);
          print(x); print(y);"
295     )
296     self.assertEqual(result.outputs, ["10 : Integer", "11 : Integer
          "])
297
298 def test_nested_function_calls(self) -> None:
299     result = run_pipeline(
300         "def add(a, b) { return a + b; } print(add(add(1, 2), 3));"
301     )
302     self.assertEqual(result.outputs, ["6 : Integer"])
303
304     # --- print() with all four types ---
305
306 def test_print_integer(self) -> None:
307     result = run_pipeline("print(42);")
308     self.assertEqual(result.outputs, ["42 : Integer"])
309
310 def test_print_float(self) -> None:
311     result = run_pipeline("print(3.14);")
312     self.assertEqual(result.outputs, ["3.14 : Float"])
313
314 def test_print_boolean(self) -> None:
315     result = run_pipeline("print(true);")
316     self.assertEqual(result.outputs, ["true : Boolean"])
317
318 def test_print_string(self) -> None:
319     result = run_pipeline('print("plc");')
320     self.assertEqual(result.outputs, ['"plc" : String'])
321
322     # --- Reassignment ---
323
324 def test_variable_can_be_reassigned_same_type(self) -> None:
325     result = run_pipeline("x = 1; x = 2; x = 3; print(x);")
326     self.assertEqual(result.outputs, ["3 : Integer"])
327
328 def test_scope_isolation_if_block(self) -> None:
329     # Variable defined inside an if-block must not be visible
330     outside it
331     with self.assertRaises(TypeError):

```

```
330         run_pipeline(  
331             "x = 1; if (x == 1) { inner = 99; } print(inner);"  
332         )  
333  
334  
335 if __name__ == "__main__":  
336     unittest.main()
```

Listing 8.11: tests/test_pipeline.py — automated regression suite (54 tests)